

```

(kali㉿kali)-[~/HTTP_Wireshark_Assignment]
$ sudo su
[sudo] password for kali:
(kali㉿kali)-[~/HTTP_Wireshark_Assignment]
# ls -l /var/www/html

ls: cannot access '-l': No such file or directory
/var/www/html:
basic.html  firewall_lab.html  image.html  index.html  index.nginx-debian.html  lab_photo.jpg

(kali㉿kali)-[~/HTTP_Wireshark_Assignment]
# chmod 777 /var/www/html

(kali㉿kali)-[~/HTTP_Wireshark_Assignment]
# exit

(kali㉿kali)-[~/HTTP_Wireshark_Assignment]
$ ls -l
total 4
-rw-rw-r-- 1 kali kali 166 Sep  8 07:35 basic.html

```

```

(kali㉿kali)-[~/HTTP_Wireshark_Assignment]
$ ls -l /var/www/html
total 84
-rw-r--r-- 1 root root 166 Sep  8 07:36 basic.html
-rw-r--r-- 1 root root 126 Aug  1 11:11 firewall_lab.html
-rw-r--r-- 1 root root 149 Jul 26 08:32 image.html
-rw-r--r-- 1 root root 205 Aug 10 19:54 index.html
-rw-r--r-- 1 root root 615 Mar 20 02:43 index.nginx-debian.html
-rw-r--r-- 1 root root 63522 Jul 26 08:31 lab_photo.jpg

(kali㉿kali)-[~/HTTP_Wireshark_Assignment]
$ mkdir -p ~/traffic

(kali㉿kali)-[~/HTTP_Wireshark_Assignment]
$ wireshark -i lo -k -w ~/traffic/image2.pcapng
** (wireshark:35202) 08:29:01.283879 [GUI WARNING] -- Failed to register with host portal QDBusError("org.freedesktop.portal.Error.Failed",
"Could not register app ID: Connection already associated with an application ID")
** (wireshark:35202) 08:29:02.273788 [Capture MESSAGE] -- Capture Start ...
** (wireshark:35202) 08:29:02.514859 [Capture MESSAGE] -- Capture started
** (wireshark:35202) 08:29:02.515951 [Capture MESSAGE] -- File: "/home/kali/traffic/image2.pcapng"
** (wireshark:35202) 08:30:33.342957 [Capture MESSAGE] -- Capture Stop ...
** (wireshark:35202) 08:30:33.386846 [Capture MESSAGE] -- Capture stopped.

```

```

(kali㉿kali)-[~]
$ mkdir -p HTTP_Wireshark_Assignment
cd HTTP_Wireshark_Assignment

(kali㉿kali)-[~/HTTP_Wireshark_Assignment]
$ nano basic.html

(kali㉿kali)-[~/HTTP_Wireshark_Assignment]
$ sudo cp basic.html /var/www/html/
[sudo] password for kali:

(kali㉿kali)-[~/HTTP_Wireshark_Assignment]
$ sudo systemctl restart apache2

(kali㉿kali)-[~/HTTP_Wireshark_Assignment]
$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-08 07:36:49 EDT; 14s ago
 Invocation: 392e851a4ae646978c8a9c1fd5b90948
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 7784 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 6 (limit: 2075)
    Memory: 18.3M (peak: 18.6M)
       CPU: 108ms

```

```
CGroup: /system.slice/apache2.service
├─7784 /usr/sbin/apache2 -k start -DFOREGROUND
├─7788 /usr/sbin/apache2 -k start -DFOREGROUND
├─7789 /usr/sbin/apache2 -k start -DFOREGROUND
├─7790 /usr/sbin/apache2 -k start -DFOREGROUND
├─7791 /usr/sbin/apache2 -k start -DFOREGROUND
└─7793 /usr/sbin/apache2 -k start -DFOREGROUND
```

Sep 08 07:36:49 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...

Sep 08 07:36:49 kali apachectl[7784]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please edit the 'ServerName' line in the default conf file to eliminate this warning.

Sep 08 07:36:49 kali systemd[1]: Started apache2.service - The Apache HTTP Server.

lines 1-21/21 (END)

HTTP Wireshark Forensics

http://127.0.0.1/basic.html

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tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
321	47.189608736	127.0.0.1	127.0.0.1	TCP	74	53268 → 80 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=3497704989 TSecr=0 WS=256
322	47.189625085	127.0.0.1	127.0.0.1	TCP	74	80 → 53268 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=3497704989 TSecr=3497704989 WS=256
323	47.189630122	127.0.0.1	127.0.0.1	HTTP	60	53268 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=3497704989 TSecr=3497704989
324	47.189672115	127.0.0.1	127.0.0.1	HTTP	149	GET /basic.html HTTP/1.1
325	47.189722540	127.0.0.1	127.0.0.1	TCP	66	80 → 53268 [ACK] Seq=1 Ack=84 Win=65536 Len=0 TSval=3497704989 TSecr=3497704989
326	47.231998716	127.0.0.1	127.0.0.1	HTTP	483	HTTP/1.1 200 OK (text/html)
327	47.232638594	127.0.0.1	127.0.0.1	TCP	66	53268 → 80 [ACK] Seq=84 Ack=418 Win=65280 Len=0 TSval=3497705031 TSecr=3497705031
328	47.232682017	127.0.0.1	127.0.0.1	TCP	66	53268 → 80 [FIN, ACK] Seq=84 Ack=418 Win=65536 Len=0 TSval=3497705032 TSecr=3497705031
329	47.234228769	127.0.0.1	127.0.0.1	TCP	66	80 → 53268 [FIN, ACK] Seq=418 Ack=85 Win=65536 Len=0 TSval=3497705034 TSecr=3497705032
330	47.234253793	127.0.0.1	127.0.0.1	TCP	66	53268 → 80 [ACK] Seq=85 Ack=419 Win=65536 Len=0 TSval=3497705034 TSecr=3497705034

Frame 323: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 53268, Dst Port: 80, Seq: 1, Ack: 1, Len: 0
Source Port: 53268
Destination Port: 80
[Stream index: 0]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 447042374
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 235143597
1000 = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 256
[Calculated window size: 65536]
[Window size scaling factor: 256]
Checksum: 0xfe28 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]

Specifies if this is an individual (unicast) or group (broadcast/multicast) address (eth.dst,lg), 1 bit

Packets: 7218 - Displayed: 10 (0.1%)

Profile: Default

No.	Time	Source	Destination	Protocol	Length	Info
324	47.189672115	127.0.0.1	127.0.0.1	HTTP	149	GET /basic.html HTTP/1.1
326	47.231998716	127.0.0.1	127.0.0.1	HTTP	483	HTTP/1.1 200 OK (text/html)

Frame 324: Packet, 149 bytes on wire (1192 bits), 149 bytes captured (1192 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
Destination: 00:00:00:00:00:00 (00:00:00:00:00:00)
... .. = IG bit: Globally unique address (factory default)
... .. = IG bit: Individual address (unicast)
Source: 00:00:00:00:00:00 (00:00:00:00:00:00)
... .. = IG bit: Globally unique address (factory default)
... .. = IG bit: Individual address (unicast)
Type: IPv4 (0x0000)
[Stream index: 0]
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 53268, Dst Port: 80, Seq: 1, Ack: 1, Len: 83
Source Port: 53268
Destination Port: 80
[Stream index: 0]
[Stream Packet Number: 4]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 83]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 447042374
[Next Sequence Number: 84 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 235143597
1000 = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)
Window: 256
[Calculated window size: 65536]
[Window size scaling factor: 256]
Checksum: 0xfe7b [unverified]

Hypertext Transfer Protocol: Protocol

Packets: 84 - Displayed: 2 (0.0%)

Profile: Default

```
Sep 08 07:36:49 kali systemd[1]: Starting apache2.service - The Apache HTTP Server
Sep 08 07:36:49 kali apache2[7784]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.0.1
Sep 08 07:36:49 kali systemd[1]: Started apache2.service - The Apache HTTP Server

(kali@kali)~[/HTTP_Wireshark_Assignment]
$ curl http://127.0.0.1/basic.html
<!DOCTYPE html>
<html>
<head>
<title>HTTP Wireshark Forensics</title>
</head>
<body>
<h1>My First Heading</h1>
<p>Name: Fuseini Imoru</p>
</body>
</html>

(kali@kali)~[/HTTP_Wireshark_Assignment]
$

(kali@kali)~[/HTTP_Wireshark_Assignment]
$ wireshark -i lo -k -w ~/traffic/basic.log
** (wireshark:10274) 07:40:44.838342 [GUI WARNING] -- Failed to register app ID: Connection already exists
** (wireshark:10274) 07:40:47.905456 [Capture MESSAGE] -- Capture started
** (wireshark:10274) 07:40:48.154600 [Capture MESSAGE] -- Error message d be saved ("~/home/kali/traffic/basic.log") could not be opened: No such file or directory
** (wireshark:10274) 07:40:50.725166 [Capture MESSAGE] -- Capture started

(kali@kali)~[/HTTP_Wireshark_Assignment]
$ mkdir -p ~/traffic

(kali@kali)~[/HTTP_Wireshark_Assignment]
$ wireshark -i lo -k -w ~/traffic/basic.pcapng
** (wireshark:10935) 07:41:40.219957 [GUI WARNING] -- Failed to register app ID: Connection already exists
** (wireshark:10935) 07:41:41.288060 [Capture MESSAGE] -- Capture started
** (wireshark:10935) 07:41:41.529270 [Capture MESSAGE] -- Capture started
** (wireshark:10935) 07:41:41.529318 [Capture MESSAGE] -- File: "/home/kali/traffic/basic.pcapng" could not be opened: No such file or directory
** (wireshark:10935) 08:02:56.944645 [Capture MESSAGE] -- Capture started
** (wireshark:10935) 08:02:56.994726 [Capture MESSAGE] -- Capture started

(kali@kali)~[/HTTP_Wireshark_Assignment]
$
```

```
(kali@kali)-[~/HTTP_Wireshark_Assignment]
└─$ mkdir -p ~/traffic

(kali@kali)-[~/HTTP_Wireshark_Assignment]
└─$ wireshark -i lo -k -w ~/traffic/image2.pcapng
** (wireshark:23351) 08:05:49.690881 [GUI WARNING] -- Failed to register with host portal QDBusError("org.freedesktop.portal.Error.Failed",
"Could not register app ID: Connection already associated with an application ID")
** (wireshark:23351) 08:05:50.630815 [Capture MESSAGE] -- Capture Start ...
** (wireshark:23351) 08:05:50.853707 [Capture MESSAGE] -- Capture started
** (wireshark:23351) 08:05:50.853860 [Capture MESSAGE] -- File: "/home/kali/traffic/image2.pcapng"
** (wireshark:23351) 08:08:06.055126 [Capture MESSAGE] -- Capture Stop ...
** (wireshark:23351) 08:08:06.094645 [Capture MESSAGE] -- Capture stopped.

(kali@kali)-[~/HTTP_Wireshark_Assignment]
└─$ curl http://127.0.0.1/basic.html
<!DOCTYPE html>
<html>
<head>
  <title>HTTP Wireshark Forensics</title>
</head>
<body>
  <h1>My First Heading</h1>
  <p>Name: Fuseini Imoru</p>
</body>
</html>

(kali@kali)-[~/HTTP_Wireshark_Assignment]
└─$
```

No.	Time	Source	Destination	Protocol	Length	Info
321	54.119747936	127.0.0.1	127.0.0.1	TCP	74	45780 → 80 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=3499436168 TSecr=0 WS=256
322	54.119764303	127.0.0.1	127.0.0.1	TCP	74	80 → 45780 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=3499436168 TSecr=3499436168 WS=2
323	54.119775955	127.0.0.1	127.0.0.1	TCP	66	45780 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=3499436168 TSecr=3499436168
324	54.119830473	127.0.0.1	127.0.0.1	HTTP	149	GET /basic.html HTTP/1.1
325	54.119892247	127.0.0.1	127.0.0.1	TCP	66	80 → 45780 [ACK] Seq=1 Ack=84 Win=65536 Len=0 TSval=3499436168 TSecr=3499436168
326	54.120369912	127.0.0.1	127.0.0.1	HTTP	483	HTTP/1.1 200 OK (text/html)
327	54.120443803	127.0.0.1	127.0.0.1	TCP	66	45780 → 80 [ACK] Seq=84 Ack=418 Win=65200 Len=0 TSval=3499436169 TSecr=3499436169
328	54.120609510	127.0.0.1	127.0.0.1	TCP	66	45780 → 80 [FIN, ACK] Seq=84 Ack=418 Win=65536 Len=0 TSval=3499436169 TSecr=3499436169
329	54.120710869	127.0.0.1	127.0.0.1	TCP	66	80 → 45780 [FIN, ACK] Seq=418 Ack=85 Win=65536 Len=0 TSval=3499436169 TSecr=3499436169
330	54.120736535	127.0.0.1	127.0.0.1	TCP	66	45780 → 80 [ACK] Seq=85 Ack=419 Win=65536 Len=0 TSval=3499436169 TSecr=3499436169
-	Frame 321: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on Interface lo, id 0					
-	Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)					
-	Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1					
-	Transmission Control Protocol, Src Port: 45780, Dst Port: 80, Seq: 0, Len: 0					
-	Source port: 45780					
-	Destination Port: 80					
-	[Stream index: 0]					
-	[Stream Packet Number: 1]					
-	Conversation completeness: Complete, WITH_DATA (31)]					
-	[TCP Segment Len: 0]					
-	Sequence number: 0 (relative sequence number)					
-	Sequence Number (raw): 1021500370					
-	Next Sequence Number: 1 (relative sequence number))					
-	Acknowledgment Number: 0					
-	Acknowledgment number (raw): 0					
-	1010 ... = Header Length: 40 bytes (10)					
-	Flags: 0x002 (SYN)					
-	Window: 65495					
-	[Calculated window size: 65495]					
-	Checksum: 0xfef30 [unverified]					
-	[Checksum Status: Unverified]					
-	Urgent Pointer: 0					
-	Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale					
-	[Timestamps]					
-	[Client Contiguous Streams: 1]					
-	[Server Contiguous Streams: 1]					

Wireshark packet capture analysis showing details for packet 322. The packet is a TCP segment from 127.0.0.1 to 127.0.0.1, port 45700. The details pane shows the source port, destination port, and various TCP options like window size, checksum, and sequence numbers. The packet length is 74 bytes.

Wireshark packet capture analysis showing details for packet 323. The packet is a TCP segment from 127.0.0.1 to 127.0.0.1, port 45700. The details pane shows the source port, destination port, and various TCP options like window size, checksum, and sequence numbers. The packet length is 66 bytes.

Terminal output showing the execution of Wireshark commands to filter and display packet details. The commands include filtering by source port and displaying specific fields like frame number, time relative, IP addresses, and TCP sequence numbers. The output shows a list of packets and their details.

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment]
# shred --help
Usage: shred [OPTION]... FILE...
Overwrite the specified FILE(s) repeatedly, in order to make it harder
for even very expensive hardware probing to recover the data.

If FILE is -, shred standard output.

Mandatory arguments to long options are mandatory for short options too.
  -f, --force
        change permissions to allow writing if necessary
  -n, --iterations=N
        overwrite N times instead of the default (3)
  --random-source=FILE
        get random bytes from FILE
  -s, --size=N
        shred this many bytes (suffixes like K, M, G accepted)
  -u
        deallocate and remove file after overwriting
  --remove[=HOW]
        like -u but give control on HOW to delete; See below
  -v, --verbose
        show details of data and metadata operations performed
  -x, --exact
        do not round file sizes up to the next full block;
        this is the default for non-regular files
  -z, --zero
        add a final overwrite with zeros to hide shredding
  --help
        display this help and exit
  --version
        output version information and exit
```

Delete FILE(s) if --remove (-u) is specified. The default is not to remove the files because it is common to operate on device files like /dev/hda, and those files usually should not be removed.

The optional HOW parameter indicates how to remove a directory entry:

- 'unlink' ⇒ use a standard unlink call.
- 'wipe' ⇒ also first obfuscate bytes in the name.
- 'wipesync' ⇒ also sync each obfuscated byte to the device.

The default mode is 'wipesync', but note it can be expensive.

CAUTION: shred assumes the file system and hardware overwrite data in place. Although this is common, many platforms operate otherwise. Also, backups and mirrors may contain unremovable copies that will let a shredded file be recovered later. See the GNU coreutils manual for details.

Report bugs to: bug-coreutils@gnu.org
 GNU coreutils home page: <https://www.gnu.org/software/coreutils/>
 General help using GNU software: <https://www.gnu.org/gethelp/>
 Full documentation <https://www.gnu.org/software/coreutils/shred>
 or available locally via: `info '(coreutils) shred invocation'`

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment]
# mkdir shred

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment]
# cd shred/

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred]
# echo hello > test.txt

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred]
# ls -l test.txt
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred]
```

```
# ls -l test.txt
```

```
-rw-r--r-- 1 root root 6 Sep  8 08:39 test.txt
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred]
```

```
# cat test.txt
```

```
hello
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred]
```

```
# xxd test.txt
```

```
00000000: 6865 6c6c 6f0a                hello.
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred]
```

```
# shred -vzn 3 test.txt
```

```
shred: -vzn: failed to open for writing: No such file or directory
```

```
shred: 3: failed to open for writing: No such file or directory
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred]
```

```
# shred -vzn 3 test.txt
```

```
shred: test.txt: pass 1/4 (random) ...
```

```
shred: test.txt: pass 2/4 (random) ...
```

```
shred: test.txt: pass 3/4 (random) ...
```

```
shred: test.txt: pass 4/4 (000000) ...
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred]
```

```
# cat test.txt
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred]
```

```
# xxd test.txt | head
```

```
00000000: 0000 0000 0000 0000 0000 0000 0000 0000  ....
```

```
00000010: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000020: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000030: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000040: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000050: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000060: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000070: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000080: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000090: 0000 0000 0000 0000 0000 0000 0000 0000 .....
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred]
# xxd test.txt | tail
```

```
00000f60: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000f70: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000f80: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000f90: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000fa0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000fb0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000fc0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000fd0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000fe0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000ff0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred]
# ls -l test.txt
```

```
-rw-r--r-- 1 root root 4096 Sep  8 08:43 test.txt
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# gdown "https://drive.google.com/uc?id=1kZrWI1_hyJEgpjiuNf_iAwdpJvQLEkKS"
Downloading ...
From: https://drive.google.com/uc?id=1kZrWI1_hyJEgpjiuNf_iAwdpJvQLEkKS
To: /home/kali/HTTP_Wireshark_Assignment/shred/rhion/DFRWS2005-RODEO.zip
100% | 3.61M/3.61M [00:02<00:00, 1.21MB/s]
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# ls
DFRWS2005-RODEO.zip 'view?usp=sharing'
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# unzip DFRWS2005-RODEO.zip
Archive: DFRWS2005-RODEO.zip
  inflating: RHINOUSB.dd
  inflating: rhino.log
  inflating: rhino2.log
  inflating: rhino3.log
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# openssl dgst -md5 rhino.log
MD5(rhino.log)= c0d0093eb1664cd7b73f3a5225ae3f30
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# openssl dgst -md5 rhino2.log
MD5(rhino2.log)= cd21eaf4acfb50f71ffff857d7968341
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# openssl dgst -md5 rhino3.log
MD5(rhino3.log)= 7e29f9d67346df25faaf18efcd95fc30
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# openssl dgst -md5 RHINOUSB.dd
MD5(RHINOUSB.dd)= 80348c58eec4c328ef1f7709adc56a54
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
#
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# fdisk -l RHINOUSB.dd
```

```
Disk RHINOUSB.dd: 247.48 MiB, 259506176 bytes, 506848 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x00000000
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# mmls RHINOUSB.dd
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# ls -lh RHINOUSB.dd
```

```
-rw-r--r-- 1 root root 248M Apr 30 2004 RHINOUSB.dd
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# xxd -l 512 RHINOUSB.dd
```

```
00000000: eb3c 906d 6b64 6f73 6673 0000 0208 0100  .<.mkdosfs.. ...
00000010: 0200 0200 00f8 f800 3e00 0800 0000 0000  ... ..>.....
00000020: e0bb 0700 0000 29d1 d992 4020 2020 2020  ... ..) ...@
00000030: 2020 2020 2020 4641 5431 3620 2020 0e1f          FAT16 ..
00000040: be5b 7cac 22c0 740b 56b4 0ebb 0700 cd10  .[|."t.V.....
00000050: 5eeb f032 e4cd 16cd 19eb fe54 6869 7320  ^..2.....This
```

```
00000060: 6973 206e 6f74 2061 2062 6f6f 7461 626c  is not a bootabl
00000070: 6520 6469 736b 2e20 2050 6c65 6173 6520  e disk. Please
00000080: 696e 7365 7274 2061 2062 6f6f 7461 626c  insert a bootabl
00000090: 6520 666c 6f70 7079 2061 6e64 0d0a 7072  e floppy and..pr
000000a0: 6573 7320 616e 7920 6b65 7920 746f 2074  ess any key to t
000000b0: 7279 2061 6761 696e 202e 2e2e 200d 0a00  ry again ... ..
000000c0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000d0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000e0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000f0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000100: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000110: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000120: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000130: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000140: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000150: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000160: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000170: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000180: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000190: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000001a0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000001b0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000001c0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000001d0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
```

```
000001e0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000001f0: 0000 0000 0000 0000 0000 0000 0000 55aa .....U.
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
```

```
# xxd -s 510 -l 2 RHINOUSB.dd
```

```
000001fe: 55aa U.
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
```

```
# mmls -v RHINOUSB.dd
```

```
tsk_img_open: Type: 0 NumImg: 1 Img1: RHINOUSB.dd
aff_open: Error determining type of file: RHINOUSB.dd
aff_open: Success
Error opening vmdk file
Error checking file signature for vhd file
tsk_img_findFiles: RHINOUSB.dd found
tsk_img_findFiles: 1 total segments found
raw_open: segment: 0 size: 259506176 max offset: 259506176 path: RHINOUSB.dd
dos_load_prim: Table Sector: 0
raw_read: byte offset: 0 len: 65536
raw_read: found in image 0 relative offset: 0 len: 65536
raw_read_segment: opening file into slot 0: RHINOUSB.dd
dos_load_prim_table: Testing FAT/NTFS conditions
load_pri:0:0 Start: 0 Size: 0 Type: 0
load_pri:0:1 Start: 0 Size: 0 Type: 0
load_pri:0:2 Start: 0 Size: 0 Type: 0
load_pri:0:3 Start: 0 Size: 0 Type: 0
dos_load_prim: No valid entries
```

```
bsd_load_table: Table Sector: 1
gpt_load_table: Sector: 1
gpt_open: Trying other sector sizes
gpt_open: Trying sector size: 512
gpt_load_table: Sector: 1
gpt_open: Trying sector size: 1024
gpt_load_table: Sector: 1
gpt_open: Trying sector size: 2048
gpt_load_table: Sector: 1
gpt_open: Trying sector size: 4096
gpt_load_table: Sector: 1
gpt_open: Trying sector size: 8192
gpt_load_table: Sector: 1
gpt_open: Trying secondary table
gpt_load_table: Sector: 506847
raw_read: byte offset: 259505664 len: 512
raw_read: found in image 0 relative offset: 259505664 len: 512
gpt_open: Trying secondary table sector size: 512
gpt_load_table: Sector: 506847
gpt_open: Trying secondary table sector size: 1024
gpt_load_table: Sector: 253423
raw_read: byte offset: 259505152 len: 1024
raw_read: found in image 0 relative offset: 259505152 len: 1024
gpt_open: Trying secondary table sector size: 2048
gpt_load_table: Sector: 126711
raw_read: byte offset: 259504128 len: 2048
raw_read: found in image 0 relative offset: 259504128 len: 2048
```

```
gpt_open: Trying secondary table sector size: 4096
gpt_load_table: Sector: 63355
raw_read: byte offset: 259502080 len: 4096
raw_read: found in image 0 relative offset: 259502080 len: 4096
gpt_open: Trying secondary table sector size: 8192
gpt_load_table: Sector: 31677
raw_read: byte offset: 259497984 len: 8192
raw_read: found in image 0 relative offset: 259497984 len: 8192
sun_load_table: Trying sector: 0
sun_load_table: Trying sector: 1
mac_load_table: Sector: 1
mac_load: Missing initial magic value
mac_open: Trying 4096-byte sector size instead of 512-byte
mac_load_table: Sector: 1
mac_load: Missing initial magic value
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# fdisk -lu RHINOUSB.dd
```

```
Disk RHINOUSB.dd: 247.48 MiB, 259506176 bytes, 506848 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x00000000
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# fsstat -b 512 -o 0 RHINOUSB.dd
```

```
FILE SYSTEM INFORMATION
```

```
File System Type: FAT16
```

```
OEM Name: mkdosfs
Volume ID: 0x4092d9d1
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: FAT16
```

```
Sectors before file system: 0
```

```
File System Layout (in sectors)
Total Range: 0 - 506847
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 248
* FAT 1: 249 - 496
* Data Area: 497 - 506847
** Root Directory: 497 - 528
** Cluster Area: 529 - 506840
** Non-clustered: 506841 - 506847
```

METADATA INFORMATION

Range: 2 - 8101622
Root Directory: 2

CONTENT INFORMATION

Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 2 - 63290

FAT CONTENTS (in sectors)

529-536 (8) → EOF
537-544 (8) → EOF

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
#
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
```

```
# fls -o 0 RHINOUSB.dd
```

```
r/r 4: gumbo1.txt
```

```
r/r 6: gumbo2.txt
```

```
v/v 8101619: $MBR
```

```
v/v 8101620: $FAT1
```

```
v/v 8101621: $FAT2
```

```
V/V 8101622: $OrphanFiles
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
```

```
# fls
```

```
Missing image name
```

```
usage: fls [-adDFlhpruvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-m dir/] [-o imgoffset] [-z ZONE] [-s seconds]
image [images] [inode]
```

```
If [inode] is not given, the root directory is used
```

```
-a: Display "." and ".." entries
```

```
-d: Display deleted entries only
```

```
-D: Display only directories
```

```
-F: Display only files
```

```
-l: Display long version (like ls -l)
```

```
-i imgtype: Format of image file (use '-i list' for supported types)
```

```
-b dev_sector_size: The size (in bytes) of the device sectors
```

```
-f fstype: File system type (use '-f list' for supported types)
```

```
-m: Display output in mactime input format with
```

```
dir/ as the actual mount point of the image
```

```
-h: Include MD5 checksum hash in mactime output
```

```
-o imgoffset: Offset into image file (in sectors)
```

```
-P pooltype: Pool container type (use '-P list' for supported types)
```

```
-B pool_volume_block: Starting block (for pool volumes only)
```

```
-S snap_id: Snapshot ID (for APFS only)
```

```
-p: Display full path for each file
```

```
-r: Recurse on directory entries
```

```
-u: Display undeleted entries only
```

```
-v: verbose output to stderr
```

```
-V: Print version
```

```
-z: Time zone of original machine (i.e. EST5EDT or GMT) (only useful with -l)
```

```
-s seconds: Time skew of original machine (in seconds) (only useful with -l & -m)
```

```
-k password: Decryption password for encrypted volumes
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
#
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# ls -lh
total 255M
-rw-r--r-- 1 root root 3.5M Jan  4 2026 DFRWS2005-RODEO.zip
-rw-r--r-- 1 root root  0 Sep  8 12:14 gumbo1_output.txt
-rw-r--r-- 1 root root 286K Apr 28 2004 rhino2.log
-rw-r--r-- 1 root root 221K Apr 28 2004 rhino3.log
-rw-r--r-- 1 root root 3.1M Apr 26 2004 rhino.log
-rw-r--r-- 1 root root 248M Apr 30 2004 RHINOUSB.dd
-rw-r--r-- 1 root root 78K Sep  8 11:35 'view?usp=sharing'

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# icat -o 0 RHINOUSB.dd 4 > gumbo1_output.txt

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# icat -o 0 RHINOUSB.dd 6 > gumbo2_output.txt

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# ls -l *.txt
-rw-r--r-- 1 root root 2815 Sep  8 12:15 gumbo1_output.txt
-rw-r--r-- 1 root root 1293 Sep  8 12:16 gumbo2_output.txt
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cat gumbo1_output.txt | head -n 5
SHRIMP AND TASSO GUMBO

Associate Food Editor: Alexis Touchet
Father: Rodney Miller, Abbeville, LA

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cat gumbo2_output.txt | head -n 5
SHRIMP AND ANDOUILLE SAUSAGE GUMBO

1/2 cup vegetable oil
1/2 cup all purpose flour
4 celery stalks, coarsely chopped

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# openssl dgst -md5 RHINOUSB.dd
MD5(RHINOUSB.dd)= 80348c58eec4c328ef1f7709adc56a54

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# dd if=RHINOUSB.dd of=RHINOUSB_My_COPY.dd
506848+0 records in
506848+0 records out
259506176 bytes (260 MB, 247 MiB) copied, 1.58304 s, 164 MB/s
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# openssl dgst -md5 RHINOUSB_My_COPY.dd
MD5(RHINOUSB_My_COPY.dd)= 80348c58eec4c328ef1f7709adc56a54

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# photorec RHINOUSB.dd
PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

zsh: suspended  photorec RHINOUSB.dd

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# photorec RHINOUSB_My_COPY.dd
PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

zsh: suspended  photorec RHINOUSB_My_COPY.dd

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
#
```

```
PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
```

```
PhotoRec is free software, and
comes with ABSOLUTELY NO WARRANTY.
```

```
Select a media and choose 'Proceed' using arrow keys:
```

```
>Disk RHINOUSB_My_COPY.dd - 259 MB / 247 MiB (RO)
```

```
>[Proceed] [ Quit ]
```

```
Note:
```

```
Disk capacity must be correctly detected for a successful recovery.
```

```
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
```

PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
<https://www.cgsecurity.org>

Disk RHINOUSB_My_COPY.dd - 259 MB / 247 MiB (R0)

Partition	Start	End	Size in sectors
Unknown	0 0 1 1021	6 60	506848 [Whole disk]
> P FAT16	0 0 1 1021	6 60	506848

>[Search] [Options] [File Opt] [Quit]
Start file recovery

PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
<https://www.cgsecurity.org>

P FAT16 0 0 1 1021 6 60 506848

To recover lost files, PhotoRec needs to know the filesystem type where the file were stored:

[ext2/ext3] ext2/ext3/ext4 filesystem
>[Other] FAT/NTFS/HFS+/ReiserFS/ ...

PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
<https://www.cgsecurity.org>

P FAT16 0 0 1 1021 6 60 506848

Please choose if all space needs to be analysed:

>[Free] Scan for files from FAT16 unallocated space only
[Whole] Extract files from whole partition

PhotoRec 7.2, Data Recovery Utility, February 2024

Please select a destination to save the recovered files to.
Do not choose to write the files to the same partition they were stored on.
Keys: **Arrow** keys to select another directory

C when the destination is correct

Q to quit

Directory /home/kali/HTTP_Wireshark_Assignment/shred/rhion

>drwxr-xr-x	0	0	4096	8-Sep-2026	12:29	.
drwxr-xr-x	0	0	4096	8-Sep-2026	09:06	..
-rw-----	0	0	3614418	4-Jan-2026	01:30	DFRWS2005-RODEO.zip
-rw-r--r--	0	0	259506176	30-Apr-2004	21:29	RHINOUSB.dd
-rw-r--r--	0	0	259506176	8-Sep-2026	12:24	RHINOUSB_My_COPY.dd
-rw-r--r--	0	0	2815	8-Sep-2026	12:15	gumbo1_output.txt
-rw-r--r--	0	0	1293	8-Sep-2026	12:16	gumbo2_output.txt
-rw-r--r--	0	0	40960	8-Sep-2026	12:29	photorec.se2
-rw-r--r--	0	0	3187907	26-Apr-2004	19:42	rhino.log
-rw-r--r--	0	0	292604	28-Apr-2004	18:50	rhino2.log
-rw-r--r--	0	0	226094	28-Apr-2004	18:50	rhino3.log
-rw-r--r--	0	0	79812	8-Sep-2026	11:35	view?usp=sharing

PhotoRec 7.2, Data Recovery Utility, February 2024

Christophe GRENIER <grenier@cgsecurity.org>

<https://www.cgsecurity.org>

Disk RHINOUSB_My_COPY.dd - 259 MB / 247 MiB (RO)

Partition	Start	End	Size in sectors
P FAT16	0 0 1 1021	6 60	506848

132 files saved in /recup_dir directory.

Recovery completed.

You are welcome to donate to support and encourage further development

<https://www.cgsecurity.org/wiki/Donation>

[Quit]

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# ls -l
total 514144
-rw-r--r-- 1 root root 3614418 Jan 4 2026 DFRWS2005-RODEO.zip
-rw-r--r-- 1 root root 2815 Sep 8 12:15 gumbo1_output.txt
-rw-r--r-- 1 root root 1293 Sep 8 12:16 gumbo2_output.txt
-rw-r--r-- 1 root root 40960 Sep 8 12:29 photorec.se2
drwxr-xr-x 2 root root 4096 Sep 8 12:38 recup_dir.1
drwxr-xr-x 2 root root 4096 Sep 8 12:38 recup_dir.2
-rw-r--r-- 1 root root 292604 Apr 28 2004 rhino2.log
-rw-r--r-- 1 root root 226094 Apr 28 2004 rhino3.log
-rw-r--r-- 1 root root 3187907 Apr 26 2004 rhino.log
-rw-r--r-- 1 root root 259506176 Apr 30 2004 RHINOUSB.dd
-rw-r--r-- 1 root root 259506176 Sep 8 12:24 RHINOUSB_My_COPY.dd
-rw-r--r-- 1 root root 79812 Sep 8 11:35 'view?usp=sharing'
```

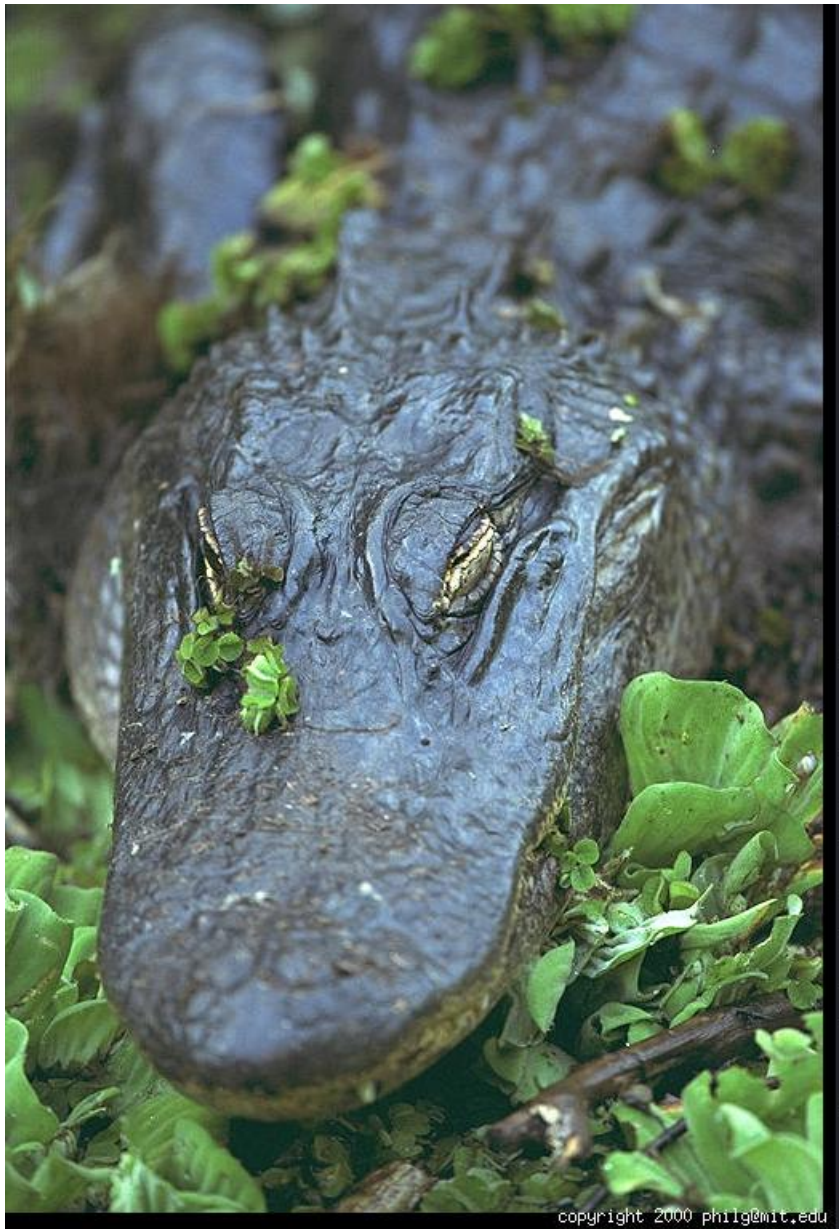
```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
#
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# ls recup_dir.2
f0000545.txt f0343137.txt f0357017.txt f0370897.txt f0384777.txt
f0104057.jpg f0343657.txt f0357537.txt f0371417.txt f0385297.txt
f0104249.jpg f0344585.txt f0358049.txt f0371929.txt f0385809.txt
f0105065.jpg f0344681.txt f0358561.txt f0372441.txt f0386321.txt
f0105873.jpg f0345201.txt f0359073.txt f0372953.txt f0386833.txt
f0106393.jpg f0345713.txt f0359593.txt f0373473.txt f0387353.txt
f0106409.jpg f0346225.txt f0360105.txt f0373985.txt f0387865.txt
f0106865.gif f0346737.txt f0360969.txt f0374497.txt f0388377.txt
f0106889.gif f0347257.txt f0361137.txt f0375009.txt f0388889.txt
f0106905.txt f0347769.txt f0361649.txt f0375529.txt f0389409.txt
f0335017_She_died_in_February_at_the_age_of_74.doc f0348281.txt f0362161.txt f0376041.txt f0389921.txt
f0335081.jpg f0348793.txt f0362673.txt f0376553.txt f0390433.txt
f0335601.txt f0349313.txt f0363193.txt f0377353.txt f0390945.txt
f0335945.txt f0349825.txt f0363705.txt f0377585.txt f0391465.txt
f0336457.txt f0350337.txt f0364217.txt f0378097.txt f0391977.txt
f0336969.txt f0350849.txt f0364729.txt f0378609.txt f0392489.txt
f0337489.txt f0351369.txt f0365249.txt f0379129.txt f0393001.txt
f0338001.txt f0351881.txt f0365761.txt f0379641.txt f0393737.txt
f0338513.txt f0352393.txt f0366273.txt f0380153.txt f0394033.txt
f0339025.txt f0352905.txt f0366785.txt f0380665.txt f0394545.txt
f0339545.txt f0353425.txt f0367305.txt f0381185.txt f0395065.txt
f0340057.txt f0353937.txt f0367817.txt f0381697.txt f0395577.txt
f0340569.txt f0354449.txt f0368329.txt f0382209.txt f0396089.txt
f0341081.txt f0354961.txt f0368841.txt f0382721.txt f0396601.txt
```

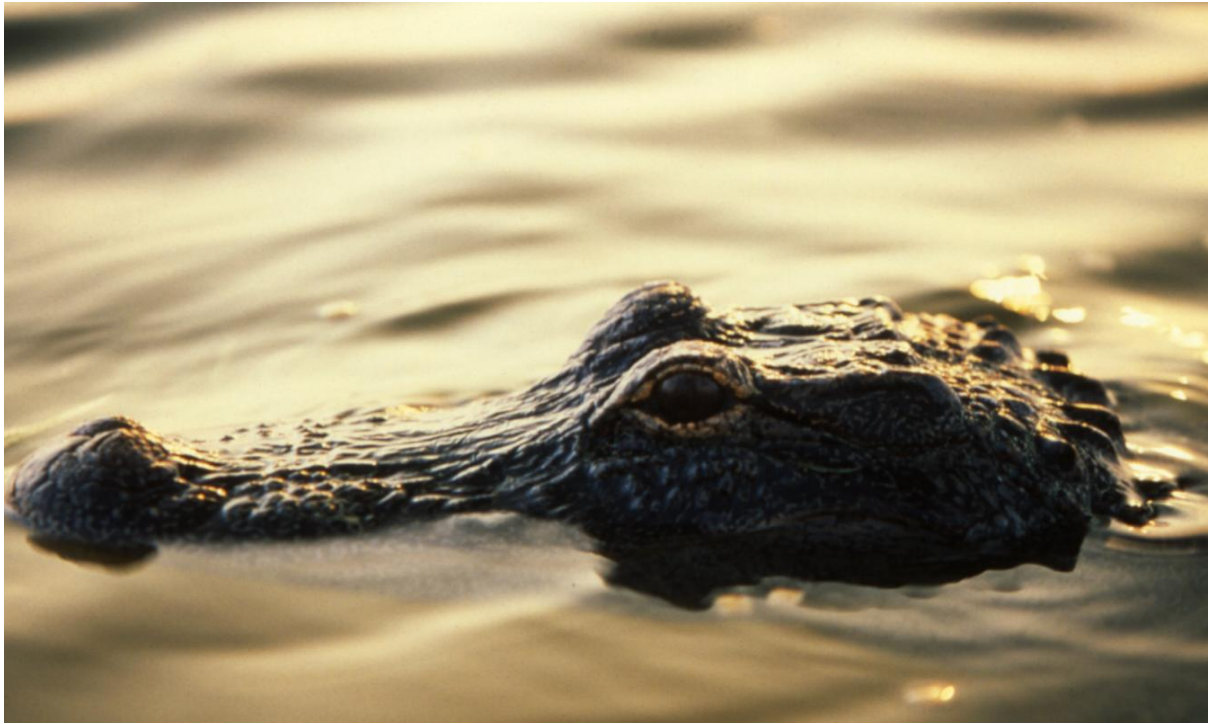
```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cd recup_dir.2

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
# ls -l *.jpg
-rw-r--r-- 1 root root 95814 Sep 8 12:38 f0104057.jpg
-rw-r--r-- 1 root root 415534 Sep 8 12:38 f0104249.jpg
-rw-r--r-- 1 root root 411361 Sep 8 12:38 f0105065.jpg
-rw-r--r-- 1 root root 264600 Sep 8 12:38 f0105873.jpg
-rw-r--r-- 1 root root 6809 Sep 8 12:38 f0106393.jpg
-rw-r--r-- 1 root root 230665 Sep 8 12:38 f0106409.jpg
-rw-r--r-- 1 root root 264600 Sep 8 12:38 f0335081.jpg
```

```
(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
#
```

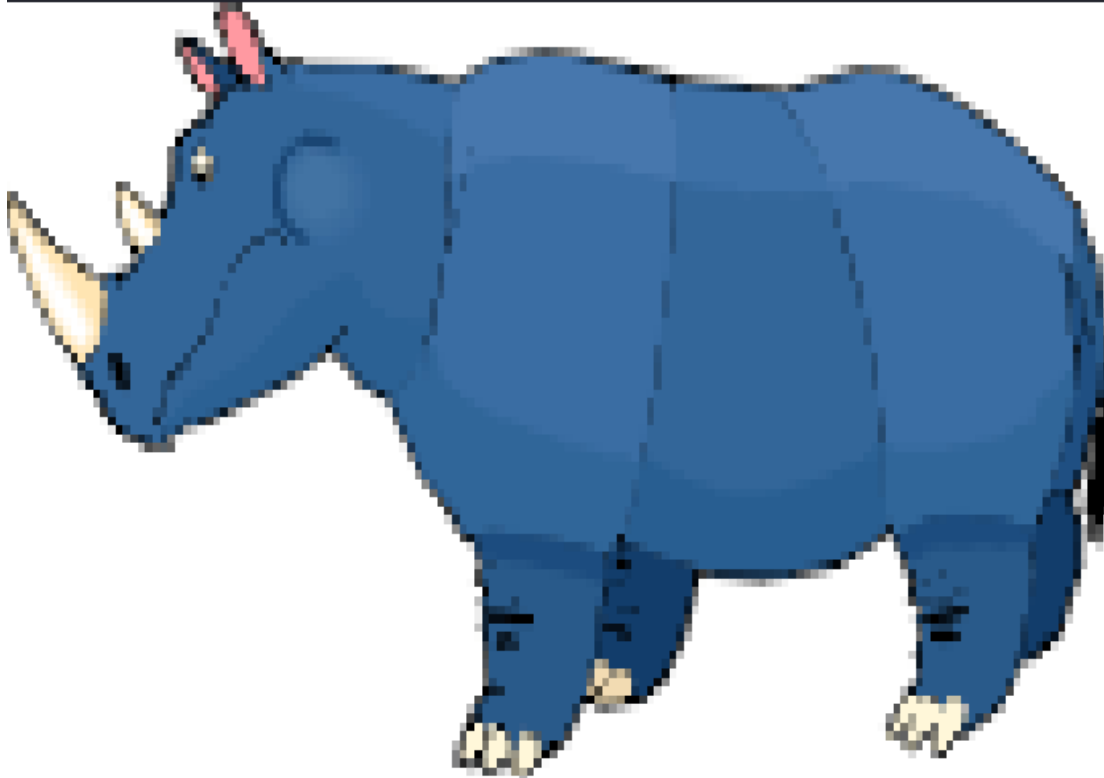
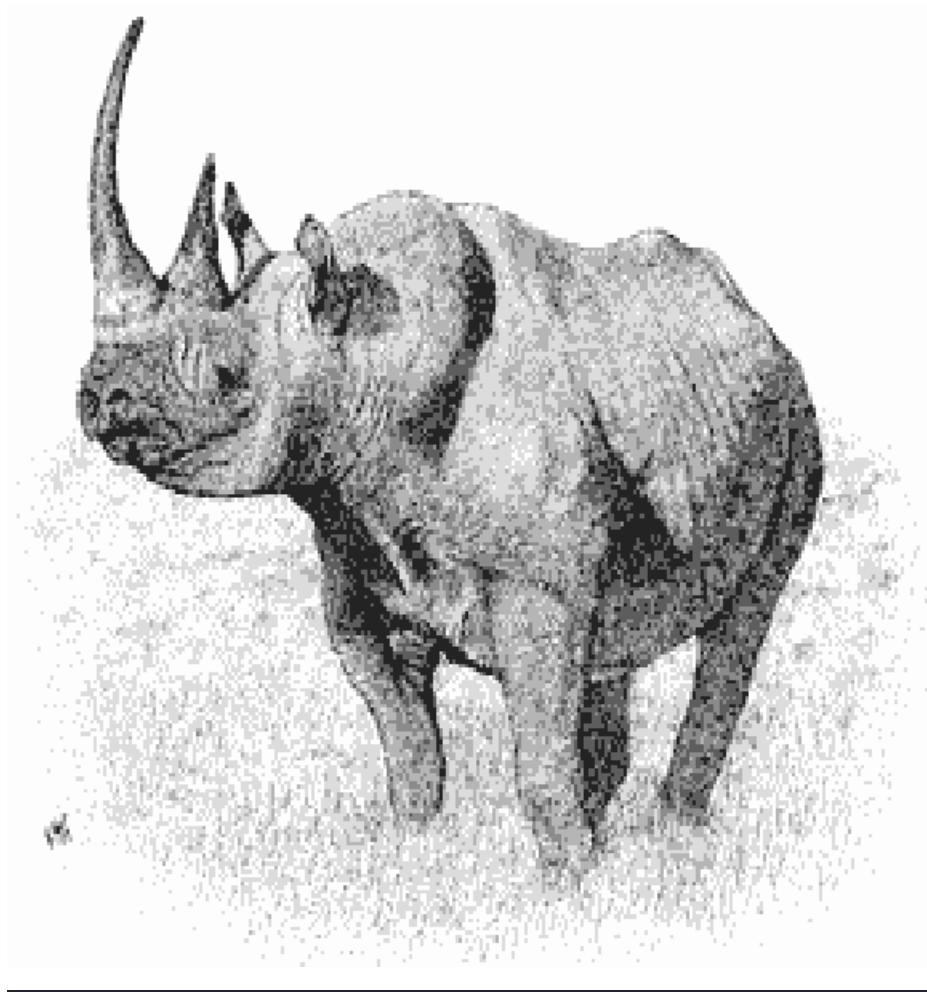


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```
(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
# display f0104057.jpg

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
# display f0104249.jpg

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
# display f0105065.jpg

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
# display f0105873.jpg

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
# display f0106393.jpg

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
# display f0106409.jpg

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
# display f0335081.jpg

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
# ls -l *.gif
-rw-r--r-- 1 root root 11407 Sep  8 12:38 f0106865.gif
-rw-r--r-- 1 root root  4105 Sep  8 12:38 f0106889.gif
```

```
(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
# display f0106865.gif

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
# display f0106889.gif

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
# ls -l *.doc
-rw-r--r-- 1 root root 30720 Aug  8 2005 f0335017_She_died_in_February_at_the_age_of_74.doc

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/recup_dir.2]
#
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cp recup_dir.2/f0104057.jpg steg/

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cp recup_dir.2/f0104249.jpg steg/

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cp recup_dir.2/f0105065.jpg steg/

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cp recup_dir.2/f0105873.jpg steg/

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cp recup_dir.2/f0106393.jpg steg/

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cp recup_dir.2/f0106409.jpg steg/

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cp recup_dir.2/f0335081.jpg steg/

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cp recup_dir.2/f0106865.gif steg/

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cp recup_dir.2/f0106889.gif steg/
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cd steg

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/steg]
# ls
f0104057.jpg  f0105065.jpg  f0106393.jpg  f0106865.gif  f0335081.jpg
f0104249.jpg  f0105873.jpg  f0106409.jpg  f0106889.gif

(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/steg]
#
```

```
(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/steg]
# exiftool f0104057.jpg
ExifTool Version Number      : 13.55
File Name                    : f0104057.jpg
Directory                    : .
File Size                    : 96 kB
File Modification Date/Time   : 2026:09:08 13:07:30-04:00
File Access Date/Time        : 2026:09:08 13:07:30-04:00
File Inode Change Date/Time   : 2026:09:08 13:07:30-04:00
File Permissions              : -rw-r--r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : None
X Resolution                  : 0
Y Resolution                  : 0
Comment                      : copyright 2000 philg@mit.edu
Image Width                  : 528
Image Height                  : 792
Encoding Process              : Baseline DCT, Huffman coding
Bits Per Sample              : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:4:4 (1 1)
Image Size                   : 528x792
Megapixels                   : 0.418
```

```
(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/steg]
# git clone https://github.com/poizan42/stegdetect

Cloning into 'stegdetect' ...
remote: Enumerating objects: 419, done.
remote: Total 419 (delta 0), reused 0 (delta 0), pack-reused 419 (from 1)
Receiving objects: 100% (419/419), 1.38 MiB | 849.00 KiB/s, done.
Resolving deltas: 100% (59/59), done.
```

```
(root@kali)-[/home/.../HTTP_Wireshark_Assignment/shred/rhion/steg]
# cd stegdetect/
```

```
(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# autoreconf -f -i
Command 'autoreconf' not found, but can be installed with:
apt install autoconf
Do you want to install it? (N/y)y
apt install autoconf
The following packages were automatically installed and are no longer required:
  libraw23t64 libvidstab1.1 openjdk-21-jre openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.
```

```
Installing:
  autoconf
```

```
Installing dependencies:
  automake autotools-dev

Suggested packages:
  autoconf-archive gnu-standards autoconf-doc libtool gettext

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 1713
  Download size: 1,454 kB
  Space needed: 4,427 kB / 790 MB available

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 autoconf all 2.73-2 [516 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 autotools-dev all 20240727.1+nmu1 [60.0 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 automake all 1:1.18.1-4 [877 kB]
Fetched 1,454 kB in 2s (672 kB/s)
Selecting previously unselected package autoconf.
(Reading database... 432984 files and directories currently installed.)
Preparing to unpack .../autoconf_2.73-2_all.deb...
Unpacking autoconf (2.73-2)...
Selecting previously unselected package autotools-dev.
Preparing to unpack .../autotools-dev_20240727.1+nmu1_all.deb...
Unpacking autotools-dev (20240727.1+nmu1)...
Selecting previously unselected package automake.
Preparing to unpack .../automake_1%3a1.18.1-4_all.deb...
Unpacking automake (1:1.18.1-4)...
Setting up autotools-dev (20240727.1+nmu1)...
```

```
└─# autoreconf -f -i
autoreconf: warning: autoconf input should be named 'configure.ac', not 'configure.in'
aclocal: warning: autoconf input should be named 'configure.ac', not 'configure.in'
aclocal: warning: autoconf input should be named 'configure.ac', not 'configure.in'
acinclude.m4:2: warning: underquoted definition of AC_C_LONG_LONG
acinclude.m4:2: run info Automake 'Extending aclocal'
acinclude.m4:2: or see https://www.gnu.org/software/automake/manual/automake.html#Extending-aclocal
configure.in:3: warning: AM_INIT_AUTOMAKE: two- and three-arguments forms are deprecated.
./lib/autoconf/general.m4:2490: AC_DIAGNOSE is expanded from...
aclocal.m4:429: AM_INIT_AUTOMAKE is expanded from...
configure.in:3: the top level
configure.in:4: warning: 'AM_CONFIG_HEADER': this macro is obsolete.
configure.in:4: You should use the 'AC_CONFIG_HEADERS' macro instead.
aclocal.m4:708: AM_CONFIG_HEADER is expanded from...
configure.in:4: the top level
configure.in:57: warning: The macro 'AC_HEADER_STDC' is obsolete.
configure.in:57: You should run autoupdate.
./lib/autoconf/headers.m4:664: AC_HEADER_STDC is expanded from...
configure.in:57: the top level
configure.in:68: warning: The macro 'AC_STRUCT_ST_RDEV' is obsolete.
configure.in:68: You should run autoupdate.
./lib/autoconf/types.m4:1058: AC_STRUCT_ST_RDEV is expanded from...
configure.in:68: the top level
configure.in:69: warning: The macro 'AC_TRY_COMPILE' is obsolete.
configure.in:69: You should run autoupdate.
./lib/autoconf/general.m4:2903: AC_TRY_COMPILE is expanded from...
lib/m4sugar/m4sh.m4:689: _AS_IF_ELSE is expanded from...
```

```

lib/m4sugar/m4sh.m4:696: AS_IF is expanded from...
./lib/autoconf/general.m4:2305: AC_CACHE_VAL is expanded from...
./lib/autoconf/general.m4:2326: AC_CACHE_CHECK is expanded from...
acinclude.m4:26: AC_STRUCT_TIMEZONE_DAYLIGHT is expanded from...
configure.in:69: the top level
configure.in:69: warning: The macro 'AC_TRY_LINK' is obsolete.
configure.in:69: You should run autoupdate.
./lib/autoconf/general.m4:2976: AC_TRY_LINK is expanded from...
lib/m4sugar/m4sh.m4:689: _AS_IF_ELSE is expanded from...
lib/m4sugar/m4sh.m4:696: AS_IF is expanded from...
./lib/autoconf/general.m4:2305: AC_CACHE_VAL is expanded from...
./lib/autoconf/general.m4:2326: AC_CACHE_CHECK is expanded from...
acinclude.m4:26: AC_STRUCT_TIMEZONE_DAYLIGHT is expanded from...
configure.in:69: the top level
configure.in:76: warning: The macro 'AC_TRY_RUN' is obsolete.
configure.in:76: You should run autoupdate.
./lib/autoconf/general.m4:3053: AC_TRY_RUN is expanded from...
lib/m4sugar/m4sh.m4:689: _AS_IF_ELSE is expanded from...
lib/m4sugar/m4sh.m4:696: AS_IF is expanded from...
./lib/autoconf/general.m4:2305: AC_CACHE_VAL is expanded from...
./lib/autoconf/general.m4:2326: AC_CACHE_CHECK is expanded from...
acinclude.m4:2: AC_C_LONG_LONG is expanded from...
configure.in:76: the top level
configure.in:85: warning: The macro 'AC_TRY_RUN' is obsolete.
configure.in:85: You should run autoupdate.
./lib/autoconf/general.m4:3053: AC_TRY_RUN is expanded from...
lib/m4sugar/m4sh.m4:689: _AS_IF_ELSE is expanded from...

```

```

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# linux32 ./configure
checking build system type... i686-pc-linux-gnu
checking host system type... i686-pc-linux-gnu
checking target system type... i686-pc-linux-gnu
checking for a BSD-compatible install... /usr/bin/install -c
checking whether sleep supports fractional seconds... yes
checking filesystem timestamp resolution... 0.01
checking whether build environment is sane... yes
checking for a race-free mkdir -p... /usr/bin/mkdir -p
checking for gawk... gawk
checking whether make sets $(MAKE)... yes
checking whether make supports nested variables... yes
checking xargs -n works... yes
checking whether UID '0' is supported by ustar format... yes
checking whether GID '0' is supported by ustar format... yes
checking how to create a ustar tar archive... gnutar
checking whether make supports the include directive... yes (GNU style)
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether gcc accepts -g... yes
checking for gcc option to enable C23 features... none needed

```

```

checking that generated files are newer than configure... done
configure: creating ./config.status
config.status: creating Makefile
config.status: creating config.h
config.status: executing depfiles commands
== configuring in jpeg-6b (/home/kali/HTTP_Wireshark_Assignment/shred/rhion/steg/stegdetect/jpeg-6b)
configure: running /bin/bash ./configure --disable-option-checking '--prefix=/usr/local' --cache-file=/dev/null --srcdir=.
checking for gcc... gcc
checking whether the C compiler (gcc ) works... no
configure: error: installation or configuration problem: C compiler cannot create executables.
configure: error: ./configure failed for jpeg-6b

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# linux32 make
make all-recursive
make[1]: Entering directory '/home/kali/HTTP_Wireshark_Assignment/shred/rhion/steg/stegdetect'
Making all in jpeg-6b
make[2]: Entering directory '/home/kali/HTTP_Wireshark_Assignment/shred/rhion/steg/stegdetect/jpeg-6b'
make[2]: *** No rule to make target 'all'. Stop.
make[2]: Leaving directory '/home/kali/HTTP_Wireshark_Assignment/shred/rhion/steg/stegdetect/jpeg-6b'
make[1]: *** [Makefile:638: all-recursive] Error 1
make[1]: Leaving directory '/home/kali/HTTP_Wireshark_Assignment/shred/rhion/steg/stegdetect'
make: *** [Makefile:455: all] Error 2

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
#

```

```

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# ./stegdetect -V
Stegdetect Version 0.6.1

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# ./stegdetect ../*.jpg
../f0104057.jpg : negative
../f0104249.jpg : jphide(*)
../f0105065.jpg : skipped (false positive likely)
../f0105873.jpg : negative
../f0106393.jpg : negative
../f0106409.jpg : negative
../f0335081.jpg : negative

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
#

```

```

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# ./stegdetect -s 3 ../*.jpg
../f0104057.jpg : negative
../f0104249.jpg : jphide(**)
../f0105065.jpg : skipped (false positive likely)
../f0105873.jpg : jphide(*)
../f0106393.jpg : negative
../f0106409.jpg : jphide(*)
../f0335081.jpg : jphide(*)

```

```
(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# find / -iname "rockyou.txt*" 2>/dev/null
/usr/share/wordlists/rockyou.txt
/usr/share/wordlists/rockyou.txt.gz

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# ./stegbreak -r rules.ini -f /usr/share/wordlists/rockyou.txt ../f0104249.jpg
Loaded 1 files ...
../f0104249.jpg : jphide[v5](gumbo)
Processed 1 files, found 1 embeddings.
Time: 8 seconds: Cracks: 131966, 16495.8 c/s

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# ./stegbreak -r rules.ini -f rockyou.txt ../f0104249.jpg
Loaded 1 files ...
stegbreak: fopen: rockyou.txt: No such file or directory

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
#
```

```
(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# ls rules.ini
rules.ini

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# mkdir /usr/local/share/stegbreak
cp rules.ini /usr/local/share/stegbreak/

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# ./stegbreak
Usage: ./stegbreak [-V] [-r <rules>] [-f <wordlist>] [-t <schemes>] file.jpg ...

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# wget -q https://github.com/brannondorsey/naive-hashcat/releases/download/data/rockyou.txt

Prepended http:// to '-q'
--2026-09-08 21:09:02-- http://xn--q-5gn/
Resolving xn--q-5gn (xn--q-5gn)... failed: Name or service not known.
wget: unable to resolve host address 'xn--q-5gn'
--2026-09-08 21:09:06-- https://github.com/brannondorsey/naive-hashcat/releases/download/data/rockyou.txt
Resolving github.com (github.com)... 140.82.121.3
Connecting to github.com (github.com)|140.82.121.3|:443 ... connected.
HTTP request sent, awaiting response... 302 Found
```

```
t%3B&filename%3Drockyou.txt&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398
a6654-997b-47e9-b12b-9515b896b4de&skt=2026-09-09T01%3A00%3A20Z&ske=2026-09-09T02%3A00%3A54Z&sks=b&skv=2018-11-09
&sig=aHrX9gGk3Zarg8xKmjn4EZ16IXxQkxOUcHgX029bzi0%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWI
uY29tIiwiaXVkiIjoicmVsZWZzZS1hc3NldHMud2Zl0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc4ODkxOTc1MCwibmJ
mIjojXzgzg4OTE2MTUwLCJwYXRoIjoicmVsZWZzZWZc2V0cHJvZHVjdGlvbi5ibG9iLmNvcuUud2luZG93cy5uZXQifQ.bwyfp529JbnduKJNjzXr
GqMxBs6H-92ONDIB-K_QWhQ&response-content-disposition=attachment%3B%20filename%3Drockyou.txt&response-content-typ
e=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.109.133, 185.19
9.108.133, 185.199.111.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.109.133|:443..
. connected.
HTTP request sent, awaiting response... 200 OK
Length: 139921497 (133M) [application/octet-stream]
Saving to: 'rockyou.txt'

rockyou.txt          100%[=====>] 133.44M  1.87MB/s   in 2m 51s

2026-09-08 21:12:17 (799 KB/s) - 'rockyou.txt' saved [139921497/139921497]

FINISHED --2026-09-08 21:12:17--
Total wall clock time: 2m 56s
Downloaded: 1 files, 133M in 2m 51s (799 KB/s)

(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
#
```

```
(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# ./stegbreak -f rockyou.txt ../*.jpg
Loaded 7 files ...
../f0105065.jpg : jphide[v5](gator)
../f0104249.jpg : jphide[v5](gumbo)
```

```
(root@kali)-[/home/.../shred/rhion/steg/stegdetect]
# mkdir stegseekexe
cd stegseekexe/

(root@kali)-[/home/.../rhion/steg/stegdetect/stegseekexe]
# wget ftp://ftp.gwdg.de/pub/linux/misc/ppdd/jphs_05.zip

--2026-09-08 21:21:23-- ftp://ftp.gwdg.de/pub/linux/misc/ppdd/jphs_05.zip
      => 'jphs_05.zip'
Resolving ftp.gwdg.de (ftp.gwdg.de)... 134.76.12.6, 2a06:93c7:f0::f79
Connecting to ftp.gwdg.de (ftp.gwdg.de)|134.76.12.6|:21... connected.
Logging in as anonymous ... Logged in!
=> SYST ... done.      => PWD ... done.
=> TYPE I ... done.    => CWD (1) /pub/linux/misc/ppdd ... done.
=> SIZE jphs_05.zip ... 184359
=> PASV ... done.      => RETR jphs_05.zip ... done.
Length: 184359 (180K) (unauthoritative)

jphs_05.zip          100%[=====] 180.04K  248KB/s  in 0.7s

2026-09-08 21:21:27 (248 KB/s) - 'jphs_05.zip' saved [184359]

(root@kali)-[/home/.../rhion/steg/stegdetect/stegseekexe]
```

```
(root@kali)-[/home/.../rhion/steg/stegdetect/stegseekexe]
# ls
jphs_05.zip

(root@kali)-[/home/.../rhion/steg/stegdetect/stegseekexe]
# unzip jphs_05.zip
Archive: jphs_05.zip
  extracting: jphs05.zip
  extracting: jphs05.zip.sig

(root@kali)-[/home/.../rhion/steg/stegdetect/stegseekexe]
# unzip jphs05.zip
Archive: jphs05.zip
  inflating: jphide.exe
  inflating: jpseek.exe
  inflating: Jphswin.exe
  inflating: Readme.txt
```

```
0050:err:vulkan:vulkan_init_once Failed to load libvulkan.so.1
0050:err:xrandr:vulkan_init_once Failed to load libvulkan.so.1
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f3025f0 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f32c9b0 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f3354e0 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f346d40 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f34fd90 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f356590 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f35d520 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f364640 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f36b5b0 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f372920 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f378db0 fd
```

Welcome to jpseek Rev 0.51
(c) 1998 Allan Latham <alatham@flexsys-group.com>
This program is freeware.
No charge is made for its use.
Use at your own risk. No liability accepted whatever happens.
Contains cryptography which may be subject to local laws.

Passphrase:

```
(root@kali)-[/home/.../rhion/steg/stegdetect/stegseekexe]
#
```



```
(root@kali)-[/home/.../rhion/steg/stegdetect/stegseekexe]
# cp ../../f0104249.jpg .
wine jpseek.exe f0104249.jpg r249.jpg
0050:err:vulkan:vulkan_init_once Failed to load libvulkan.so.1
0050:err:xrandr:vulkan_init_once Failed to load libvulkan.so.1
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f3025f0 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f32c9b0 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f335570 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f346d50 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f34fe20 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f356620 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f35d5b0 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f3645e0 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f36b550 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f3728c0 fd
00d4:err:hid:stop_polling_device could not find poll entry matching device 0x7f378d50 fd

Welcome to jpseek Rev 0.51
(c) 1998 Allan Latham <alatham@flexsys-group.com>
This program is freeware.
No charge is made for its use.
Use at your own risk. No liability accepted whatever happens.
Contains cryptography which may be subject to local laws.

Passphrase:
```



```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# git clone https://github.com/DominicBreuker/stego-toolkit.git
Cloning into 'stego-toolkit' ...
remote: Enumerating objects: 242, done.
remote: Total 242 (delta 0), reused 0 (delta 0), pack-reused 242 (from 1)
Receiving objects: 100% (242/242), 13.03 MiB | 2.64 MiB/s, done.
Resolving deltas: 100% (117/117), done.

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# ls
DFRWS2005-RODE0.zip  photorec.se2  rhino2.log  RHINOUSB.dd  stego-toolkit
gumbo1_output.txt   recup_dir.1   rhino3.log  RHINOUSB_My_COPY.dd  'view?usp=sharing'
gumbo2_output.txt   recup_dir.2   rhino.log   steg
```

```

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# cd stego-toolkit/install

(root@kali)-[/home/.../shred/rhion/stego-toolkit/install]
# chmod +x jphide.sh

(root@kali)-[/home/.../shred/rhion/stego-toolkit/install]
# ./jphide.sh
--2026-09-08 22:07:42-- https://github.com/mmayfield1/SSAK/raw/master/programs/64/jphide
Resolving github.com (github.com)... 140.82.121.4
Connecting to github.com (github.com)|140.82.121.4|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/mmtechnodrone/SSAK/raw/master/programs/64/jphide [following]
--2026-09-08 22:07:43-- https://github.com/mmtechnodrone/SSAK/raw/master/programs/64/jphide
Reusing existing connection to github.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/mmtechnodrone/SSAK/master/programs/64/jphide [following]
--2026-09-08 22:07:43-- https://raw.githubusercontent.com/mmtechnodrone/SSAK/master/programs/64/jphide
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.110.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 827960 (809K) [application/octet-stream]
Saving to: '/usr/bin/jphide'

```

```

Saving to: '/usr/bin/jpseek'

/usr/bin/jpseek          100%[=====>] 768.59K  3.17MB/s   in 0.2s

2026-09-08 22:07:47 (3.17 MB/s) - '/usr/bin/jpseek' saved [787040/787040]

(root@kali)-[/home/.../shred/rhion/stego-toolkit/install]
# jphide

jphide, version 0.3 (c) 1998 Allan Latham <alatham@flexsys-group.com>

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NO WARRANTY whatsoever is offered with this product.
NO LIABILITY whatsoever is accepted for its use.
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See the GNU Public Licence for full details.

Usage:

jphide input-jpg-file output-jpg-file hide-file

(root@kali)-[/home/.../shred/rhion/stego-toolkit/install]
#

```

```
(root@kali)-[/home/.../shred/rhion/stego-toolkit/install]
# ls -l /usr/bin/jphide
-rwxr-xr-x 1 root root 827960 Sep  8 22:07 /usr/bin/jphide

(root@kali)-[/home/.../shred/rhion/stego-toolkit/install]
# ls -l /usr/bin/jpseek
-rwxr-xr-x 1 root root 787040 Sep  8 22:07 /usr/bin/jpseek

(root@kali)-[/home/.../shred/rhion/stego-toolkit/install]
# jpseek

jpseek, version 0.3 (c) 1998 Allan Latham <alatham@flexsys-group.com>

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NO LIABILITY whatsoever is accepted for its use.
You are using this entirely at your OWN RISK.
See the GNU Public Licence for full details.

Usage:

jpseek input-file seek-file
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# ls
DFRWS2005-RODEO.zip  photorec.se2  rhino2.log  RHINOUSB.dd  stego-toolkit
gumbo1_output.txt   recup_dir.1   rhino3.log  RHINOUSB_My_COPY.dd  'view?usp=sharing'
gumbo2_output.txt   recup_dir.2   rhino.log   steg

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# ls -l
total 514152
-rw-r--r-- 1 root root 3614418 Jan  4 2026 DFRWS2005-RODEO.zip
-rw-r--r-- 1 root root 2815 Sep  8 12:15 gumbo1_output.txt
-rw-r--r-- 1 root root 1293 Sep  8 12:16 gumbo2_output.txt
-rw-r--r-- 1 root root 40960 Sep  8 12:29 photorec.se2
drwxr-xr-x 2 root root 4096 Sep  8 12:38 recup_dir.1
drwxr-xr-x 2 root root 4096 Sep  8 12:38 recup_dir.2
-rw-r--r-- 1 root root 292604 Apr 28 2004 rhino2.log
-rw-r--r-- 1 root root 226094 Apr 28 2004 rhino3.log
-rw-r--r-- 1 root root 3187907 Apr 26 2004 rhino.log
-rw-r--r-- 1 root root 259506176 Apr 30 2004 RHINOUSB.dd
-rw-r--r-- 1 root root 259506176 Sep  8 12:24 RHINOUSB_My_COPY.dd
drwxr-xr-x 3 root root 4096 Sep  8 13:23 steg
drwxr-xr-x 9 root root 4096 Sep  8 22:04 stego-toolkit
-rw-r--r-- 1 root root 79812 Sep  8 11:35 'view?usp=sharing'
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	137.30.122.253	64.233.167.104	TCP	62	1583 → 80 [SYN] Seq=0 W
2	0.082001	64.233.167.104	137.30.122.253	TCP	60	80 → 1583 [SYN, ACK] Se
3	0.082151	137.30.122.253	64.233.167.104	TCP	54	1583 → 80 [ACK] Seq=1 A
4	0.084413	137.30.122.253	64.233.167.104	HTTP	351	GET / HTTP/1.1
5	0.172955	64.233.167.104	137.30.122.253	TCP	1484	80 → 1583 [ACK] Seq=1 A
6	0.318901	137.30.122.253	64.233.167.104	TCP	54	1583 → 80 [ACK] Seq=298
7	0.396683	64.233.167.104	137.30.122.253	TCP	272	[TCP Previous segment n
8	0.396823	137.30.122.253	64.233.167.104	TCP	54	[TCP Dup ACK 6#1] 1583
9	0.397383	64.233.167.104	137.30.122.253	HTTP	1484	[TCP Out-Of-Order] HTTP
10	0.397583	137.30.122.253	64.233.167.104	TCP	54	1583 → 80 [ACK] Seq=298
11	4.071502	158.94.89.167	137.30.120.113	TCP	62	3288 → 2745 [SYN] Seq=0
12	5.377507	146.151.55.59	137.30.122.253	TCP	62	2358 → 2745 [SYN] Seq=0
13	5.831955	137.30.122.253	207.68.173.245	TCP	62	1585 → 80 [SYN] Seq=0 W
14	6.047759	207.68.173.245	137.30.122.253	TCP	62	80 → 1585 [SYN, ACK] Se
15	6.047905	137.30.122.253	207.68.173.245	TCP	54	1585 → 80 [ACK] Seq=1 A

Frame 1: Packet, 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface 0
 Ethernet II, Src: Apple_cc:57:92 (00:03:93:cc:57:92), Dst: 08:00:27:00:00:00
 Internet Protocol Version 4, Src: 137.30.122.253, Destination: 64.233.167.104
 Transmission Control Protocol, Src Port: 1583, Dst Port: 80

Packet N	Protocol	Username	Additional Info
1536	FTP	gnome	Username in packet: 1532
1629	FTP	gnome	Username in packet: 1625
5637	FTP	gnome	Username in packet: 5633

Close

No.	Time	Source	Destination	Protocol	Length	Info
1529	179.040214	137.30.120.40	137.30.122.253	FTP	82	Response: 220 cook FTP
1530	179.189473	137.30.122.253	137.30.120.40	TCP	54	1655 → 21 [ACK] Seq=1 A
1531	182.548193	128.138.182.127	137.30.123.164	TCP	62	4675 → 2745 [SYN] Seq=0
1532	182.640647	137.30.122.253	137.30.120.40	FTP	66	Request: USER gnome
1533	182.640847	137.30.120.40	137.30.122.253	TCP	60	21 → 1655 [ACK] Seq=29
1534	182.644970	137.30.120.40	137.30.122.253	FTP	88	Response: 331 Password
1535	182.800991	137.30.122.253	137.30.120.40	TCP	54	1655 → 21 [ACK] Seq=13
1536	184.667754	137.30.122.253	137.30.120.40	FTP	69	Request: PASS gnome123
1537	184.667952	137.30.120.40	137.30.122.253	TCP	60	21 → 1655 [ACK] Seq=63
1538	184.748946	137.30.120.40	137.30.122.253	FTP	81	Response: 230 User gnom
1539	184.907708	137.30.122.253	137.30.120.40	TCP	54	1655 → 21 [ACK] Seq=28
1540	185.602553	137.30.122.253	137.30.120.40	FTP	62	Request: TYPE I
1541	185.602818	137.30.120.40	137.30.122.253	FTP	74	Response: 200 Type set
1542	185.710258	137.30.122.253	137.30.120.40	TCP	54	1655 → 21 [ACK] Seq=36

▶ Frame 1536: Packet, 69 bytes on wire (552 bits), 69 bytes captured (552 bits) on interface 0
 ▶ Ethernet II, Src: Apple_cc:57:92 (00:03:93:cc:57:92), Dst: 08:00:20:f0:13:96
 ▶ Internet Protocol Version 4, Src: 137.30.122.253, Dst: 137.30.120.40
 ▶ Transmission Control Protocol, Src Port: 1655, Dst Port: 21
 ▶ File Transfer Protocol (FTP)
 Request command: PASS
 Request arg: gnome123
 [Current working directory:]

220 cook FTP server ready.

USER gnome

331 Password required for gnome.

PASS gnome123

230 User gnome logged in.

TYPE I

200 Type set to I.

PORT 137,30,122,253,6,121

200 PORT command successful.

STOR rhino1.jpg

150 Opening BINARY mode data connection for rhino1.jpg.

226 Transfer complete.

QUIT

221-You have transferred 65703 bytes in 1 files.

221-Total traffic for this session was 66042 bytes in 1 transfers.

221-Thank you for using the FTP service on cook.

221 Goodbye.

6 client pkts, 9 server pkts, 12 turns.

Entire conversation (487 bytes)

Show as ASCII

No delta times

Stream 69

Find:

Case sensitive

Find Next

Help

Filter Out This Stream

Print

Save as...

Back

Close

No.	Time	Source	Destination	Protocol	Length	Info
1539	184.907708	137.30.122.253	137.30.120.40	TCP	54	1655 → 21 [ACK] Seq=28 Ack=90 Win=64151 Len=0
1540	185.602553	137.30.122.253	137.30.120.40	FTP	62	Request: TYPE I
1541	185.602818	137.30.120.40	137.30.122.253	FTP	74	Response: 200 Type set to I.
1542	185.710258	137.30.122.253	137.30.120.40	TCP	54	1655 → 21 [ACK] Seq=36 Ack=110 Win=64131 Len=0
1544	188.994914	137.30.122.253	137.30.120.40	FTP	81	Request: PORT 137,30,122,253,6,121
1545	188.995539	137.30.120.40	137.30.122.253	FTP	84	Response: 200 PORT command successful.
1546	188.996001	137.30.122.253	137.30.120.40	FTP	71	Request: STOR rhino1.jpg
1550	189.833465	137.30.120.40	137.30.122.253	FTP	111	Response: 150 Opening BINARY mode data connection for rhino1.jpg.
1611	189.221464	137.30.122.253	137.30.120.40	TCP	54	1655 → 21 [ACK] Seq=80 Ack=197 Win=64044 Len=0
1612	189.221711	137.30.120.40	137.30.122.253	FTP	78	Response: 226 Transfer complete.
1613	189.422115	137.30.122.253	137.30.120.40	TCP	54	1655 → 21 [ACK] Seq=80 Ack=221 Win=64020 Len=0
1614	194.426879	137.30.122.253	137.30.120.40	FTP	60	Request: QUIT
1615	194.427484	137.30.120.40	137.30.122.253	FTP	104	Response: 221-You have transferred 65703 bytes in 1 files.
1616	194.432107	137.30.120.40	137.30.122.253	FTP	100	Response: 221-Total traffic for this session was 66042 bytes in 1 transfers.
1617	194.432192	137.30.122.253	137.30.120.40	TCP	54	1655 → 21 [ACK] Seq=86 Ack=404 Win=63838 Len=0

Frame 1550: Packet, 111 bytes on wire (888 bits), 111 bytes captured (888 bits) on interface 0
 Ethernet II, Src: Apple_cc:57:92 (00:03:93:cc:57:92), Dst: 08:00:20:f0:13:96
 Internet Protocol Version 4, Src: 137.30.122.253, Dst: 137.30.120.40
 Transmission Control Protocol, Src Port: 1655, Dst Port: 21
 File Transfer Protocol (FTP)
 Response: 150 Opening BINARY mode data connection for rhino1.jpg.

[illegible]

Entire conversation (65 kB) Show as Raw No delta times Stream 71
Find: ☐ Case sensitive Find Next
 Help Filter Out This Stream Print Save as... Back × Close

Description	Extension	Magic Number
Adobe Illustrator	.ai	25 50 44 46 [%PDF]
Bitmap graphic	.bmp	42 4D [BM]
Class File	.class	CA FE BA BE
JPEG graphic file	.jpg	FFD8
JPEG 2000 graphic file	.jp2	0000000C6A502020D0A [...JP..]
GIF graphic file	.gif	47 49 46 38 [GIF89]
TIF graphic file	.tif	49 49 [II]
PNG graphic file	.png	89 50 4E 47 .PNG
WAV audio file	.wav	52 49 46 46 RIFF
ELF Linux EXE	.	7F 45 4C 46 .ELF
Photoshop Graphics	.psd	38 42 50 53 [8BPS]
Windows Meta File	.wmf	D7 CD C6 9A
MIDI file	.mid	4D 54 68 64 [MThd]
Icon file	.ico	00 00 01 00
MP3 file with ID3 identity tag	.mp3	49 44 33 [ID3]
AVI video file	.avi	52 49 46 46 [RIFF]
Flash Shockwave	.swf	46 57 53 [FWS]
Flash Video	.flv	46 4C 56 [FLV]
Mpeg 4 video file	.mp4	00 00 00 18 66 74 79 70 6D 70 34 32 [...ftypmp42]
MOV video file	.mov	6D 6F 6F 76 [...moov]
Windows Video file	.wmv	30 26 B2 75 8E 66 CF
Windows Audio file	.wma	30 26 B2 75 8E 66 CF
PKZip	.zip	50 4B 03 04 [PK]
GZip	.gz	1F 8B 08

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# xxd rhino1.jpg | head
00000000: ffd8 ffe0 0010 4a46 4946 0001 0201 0048 .....JFIF.....H
00000010: 0048 0000 ffed 1598 5068 6f74 6f73 686f .H.....Photosho
00000020: 7020 332e 3000 3842 494d 03e9 0000 0000 p 3.0.8BIM.....
00000030: 0078 0003 0000 0048 0048 0000 0000 02d8 .x.....H.H.....
00000040: 0228 ffe1 ffe2 02f9 0246 0347 0528 03fc .(.....F.G.(..
00000050: 0002 0000 0048 0048 0000 0000 02d8 0228 .....H.H.....(
00000060: 0001 0000 0064 0000 0001 0003 0303 0000 .....d.....
00000070: 0001 270f 0001 0001 0000 0000 0000 0000 ..'.....
00000080: 0000 0000 6008 0019 0190 0000 0000 0000 ....`.....
00000090: 0000 0000 0000 0000 0000 0000 0000 0000 .....

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# xxd rhino1.jpg | tail
00010010: e4a4 4c63 bb51 8eeb ece4 8a38 99de 40d2 ..Lc.Q.....8..@.
00010020: 1af1 53b0 ff00 338c 81ff 0035 808e fba8 ..S...3....5....
00010030: 1b97 92a6 48fd 3463 4046 e001 f2fe 6c44 ....H.4c@F....lD
00010040: 6b70 520a 1af5 add8 a395 1cf7 0b4c 30b0 kpR.....L0.
00010050: 896e ddb9 e3e9 b722 b2f2 fdd8 eff4 6326 .n.....".....c&
00010060: ce7c d137 f2fa 93c1 15c4 214c 6e1b 915f .|.7.....!Ln.._
00010070: b5ca 9f68 fcc6 434e 68d8 3e94 4c1e 5fc4 ...h..CNh.>.L._.
00010080: a7e6 b956 6d50 b0e3 411a 8017 a0f1 fd79 ...VmP..A.....y
00010090: 2d3f 23fd 6639 4557 f552 2f56 4fb3 5f87 -?#.f9EW.R/VO._.
000100a0: c332 a834 dbff d9 .2.4 ...
```



tcp.stream eq 79

Time	Source
1655 2004-04-26 22:22:08.9425612	137.39.12
1656 2004-04-26 22:22:08.9431202	137.39.12
1657 2004-04-26 22:22:08.9432362	137.39.12
1658 2004-04-26 22:22:08.9433142	137.39.12
1659 2004-04-26 22:22:08.9443322	137.39.12
1660 2004-04-26 22:22:08.9444202	137.39.12
1661 2004-04-26 22:22:08.9444902	137.39.12
1662 2004-04-26 22:22:08.9445832	137.39.12
1663 2004-04-26 22:22:08.9445892	137.39.12
1664 2004-04-26 22:22:08.9446682	137.39.12
1665 2004-04-26 22:22:08.9447462	137.39.12
1666 2004-04-26 22:22:08.9455482	137.39.12
1667 2004-04-26 22:22:08.9456352	137.39.12
1668 2004-04-26 22:22:08.9457152	137.39.12
1669 2004-04-26 22:22:08.9457932	137.39.12

Frame 1662: Packet: 80 bytes on wire (488 bits)
Ethernet II, Src: Oracle 10:13:96 (08:00:20:10:13:96), Dst: Oracle 10:13:96 (08:00:20:10:13:96)
Internet Protocol Version 4, Src: 137.39.120, Dst: 137.39.120
Transmission Control Protocol, Src Port: 20, Dst Port: 20

Entire conversation (96 kB)

Show as Raw

No delta times

Stream: 79

Find:

Case sensitive

Find Next




```
b57fe628eef8a9fdd17cb32a73da1368bfbdb64c6f44b8b5b7deef76ba645974baa84e8a4195c3b64ebf48c411
8fe69f96e0449fea908f620533ff103ccf334776985c23832b416bd5f8cbb8f0aec3455faeab24ac746713813
b2c9a778414f4d1af9b9c2d53a1f9807fc5fb726bb0e2caa3616cee361e1bc166788bb13befa3b7c54d30911b
4adeff52c2bce8efbdc0e8c7ebd879e7bd10170e7244b0d0230e500858a816fe2edf4f87ab1297d05301287c
37e2ebac8b023babf02d7100b5d3183866761f60163526d81840a16ff85ac08851ef308e63972a828bcf10bb1
5c870f24433b25f77cb2149405bf3904e9cfc2ef13a93a29c7a57b24b3d1a7c38259433621ec8193087d394f6
f474db065a06849e6cf4617e8d777f2003eebfe1cb69a4e8be27096b524f6810241a9d70e17452766faf3ea33
1e921be32e40ca8d5294fb3e12d5bc99517f41d08da6a7f8a3b07526befb76ea6ac6aa90515fba619512eeafb
9934ceee30c0d67972a83ba38a147b31d43ab985083e3fd64b199338a35105204dfc4bbfe66c41cf10a7bab0e
1e70c9f2338b7a3d74fe544063282e8fab5a17689fddfbb9de8c8dd54f7f29c9a64e2672c3e7f2f4c2d92b1b20
bf473d5c554542224680d2edc1f0dcc15fd745e048e792e4c69f65a00ff2d68c95e6645f93ea7b7843ff383db
820d7c5036d417a69d35960cba727dd92171e14baaf6f0861734a6808f177a54630c8694a990e919136475771
bea71aa31ba20b8d0f08037821612eb5b39000d252750a3f13e1d5f736e353f7dc3f42d5b0b617b1e31fd16af
33b645b35878039fc3f7df6534d3813b612ed26c23a824e78e0d55279d6c0e1a784cd1ae048911a4c05f383a9
c346d297fd2892bbebccc1cfe77297896399b92698f968f07f69a8903123ec19e2bc86ca8245170cc0747ce2c
e84f6708aca689e22bf0916378e1f0d86079d693e07409593881184da15373e028b0135bc1d836310c8fd7787
8537b53d7c4da6e62d380997423750b0b3545b517f8a8b176c9524be6b3fb8349b69c5a7f6cf2b19d46b8d723
0071712704ad58e7b4f7139e13accc193e197dc4bb67d3650f0850c6a7e9196770974d2bd5562d540bf2ad132
b67bec3a0c74c6be00d22afe4bfddb3f5afb932e0dd3047c92af676039317151311a4f7b18ba15a0d86d3533
04d7e85b8ada1fdd4808b488b9a9e55603bc78664aed1194b59895e1ab7d380696dfa16d355e82f36e6aba42a
92b07fbc24eac1ecccc44e74647399d3fe7bae57936963104fb34ded34c7d1a3a746c3b7a56977d36620b7b40
066db18a7e60f901e50ca9c2059b003ae35b1e7605fff7945dd97a133e97d9d536235d0c825ce07e13bbf1d20
abcd021243dece8bc733d2239b24dd67b1cdcccd181fd114d9e6cc63757fc5346cadd0c3764bf1e0ebaad559c
7cfff6f6c600a33e9a641b4e01f9b8cd5e8b77a91806905b4f3b8fc1f4a61bfa105e0d17c53e7d5d0c95d53290
8219deb61393c4a0ef54f22c7458d50fa10387af040e3803b338b6bb9955b6ac88914236774baeab617a263e
99b2e62b55770f9c5e37c8c223883d90c7da2e882c7ff82c2b890d1934809fd87839c8bdec81909a1cf695c47
9a6f7b68c48ccf2469f3fa94671f96ed278e90d0e20f8e11ea9968c96659901c900e60339d1e07488d58cf
389d9ca80d5588ab66603ea9ce4e4df942799e23d159504b0102140014000100080006889a3065be6e9330840
300098503000a0000000000000000000000000000000000000000000000000000000000000000000000000000
0038000000588403000000
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# ls -l contraband.zip
-rw-r--r-- 1 root root 230566 Sep  9 06:30 contraband.zip
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# xxd contraband.zip | head
00000000: 504b 0304 1400 0100 0800 0688 9a30 65be  PK.....0e.
00000010: 6e93 3084 0300 0985 0300 0a00 0000 7268  n.0.....rh
00000020: 696e 6f32 2e6a 7067 c97e 43ad a105 2f94  ino2.jpg.~C.../.
00000030: cb18 8af6 1b1e f51e 89fb 6fcf 71c7 1e99  .....o.q...
00000040: 69fa 9d19 7d17 2ebd 4b22 6f4c c482 2e2d  i...}...K"oL...-
00000050: d3b6 9559 fcaa b0a5 3315 bc7e cb63 510f  ...Y....3..~.cQ.
00000060: 6206 506d 1c0f e621 952c df42 b966 5918  b.Pm...!.,.B.fY.
00000070: 1b04 5956 bf7d e875 7d68 d9ac 84f8 bf1b  ..YV.}.u}h.....
00000080: 257d 58e1 49fc a5b8 22eb 91d6 7da9 667f  %}X.I... " ...}.f.
00000090: 330c ce4b 5ad8 3d8e c0ce 9002 cfe2 c954  3..KZ.=.....T
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# xxd contraband.zip | tail
00038410: 9a1c f695 c479 a6f7 96f6 8c48 cecf 2469  ....y.....H.$i
00038420: f3fa 9467 1f96 ed27 8e90 d0e2 0f8e 11ea  ...g...'.....
00038430: 9968 c966 5990 1c90 0e60 3391 de07 488d  .h.fY....`3...H.
00038440: 58cf 389d 9ca8 0d55 88ab 6660 3ea9 ce4e  X.8....U..f`>..N
00038450: 4df9 4279 9e23 d159 504b 0102 1400 1400  M.By.#.YPK.....
00038460: 0100 0800 0688 9a30 65be 6e93 3084 0300  ....0e.n.0...
00038470: 0985 0300 0a00 0000 0000 0000 0000 2000  ....
00038480: 0000 0000 0000 7268 696e 6f32 2e6a 7067  ....rhino2.jpg
00038490: 504b 0506 0000 0000 0100 0100 3800 0000  PK.....8...
000384a0: 5884 0300 0000  X....
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
#
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# unzip contraband.zip
Archive:  contraband.zip
[contraband.zip] rhino2.jpg password:
zsh: suspended  unzip contraband.zip

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# apt -get install fcrackzip
E: Command line option 'g' [from -get] is not understood in combination with the other options.

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# apt-get install fcrackzip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following packages were automatically installed and are no longer required:
  libmbcrypto16 libpython3.13-dev libraw23t64 libvidstab1.1 openjdk-21-jre openjdk-21-jre-headless
  python3.13 python3.13-dev python3.13-minimal python3.13-venv rpcsvc-proto
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  fcrackzip
0 upgraded, 1 newly installed, 0 to remove and 1625 not upgraded.
Need to get 27.5 kB of archives.
After this operation, 81.9 kB of additional disk space will be used.
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# fcrackzip --help

fcrackzip version 1.0, a fast/free zip password cracker
written by Marc Lehmann <pcg@goof.com> You can find more info on
http://www.goof.com/pcg/marc/

USAGE: fcrackzip
      [-b|--brute-force]      use brute force algorithm
      [-D|--dictionary]      use a dictionary
      [-B|--benchmark]      execute a small benchmark
      [-c|--charset charset] use characters from charset
      [-h|--help]            show this message
      [--version]            show the version of this program
      [-V|--validate]        sanity-check the algorithm
      [-v|--verbose]         be more verbose
      [-p|--init-password string] use string as initial password/file
      [-l|--length min-max]  check password with length min to max
      [-u|--use-unzip]        use unzip to weed out wrong passwords
      [-m|--method num]      use method number "num" (see below)
      [-2|--modulo r/m]      only calculate 1/m of the password
                             file... the zipfiles to crack
```

```
      [-m|--method num]      use method number "num" (see below)
      [-2|--modulo r/m]      only calculate 1/m of the password
                             file... the zipfiles to crack

methods compiled in (* = default):

0: cpmask
1: zip1
*2: zip2, USE_MULT_TAB

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# wget -q https://github.com/brannondorsey/naive-hashcat/releases/download/data/rockyou.txt
```

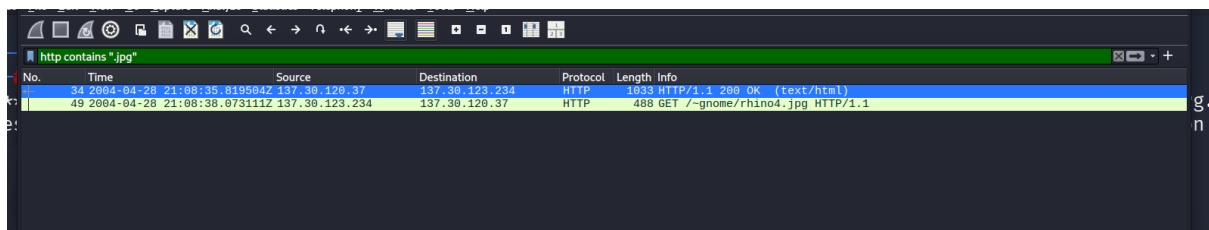
```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# fcrackzip -u -D -p rockyou.txt contraband.zip
```

PASSWORD FOUND!!!!: pw = monkey

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# unzip contraband.zip
Archive:  contraband.zip
[contraband.zip] rhino2.jpg password:
  inflating: rhino2.jpg
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# ls -l rhino2.jpg
-rw-r--r-- 1 root root 230665 Apr 26  2004 rhino2.jpg
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# display rhino2.jpg
```



```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# wireshark rhino2.log
** (wireshark:144387) 07:29:46.806178 [GUI WARNING] -- Failed to register with host portal QDBusError("org.free
desktop.portal.Error.Failed", "Could not register app ID: Connection already associated with an application ID")

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# whois 137.30.123.234

#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

NetRange:      137.30.0.0 - 137.30.255.255
CIDR:          137.30.0.0/16
NetName:       UNO-NET-BIG
NetHandle:     NET-137-30-0-1
Parent:        NET137 (NET-137-0-0-0)
```

```
NetType:       Direct Allocation
OriginAS:
Organization:  University of New Orleans (UNO)
RegDate:       1989-10-17
Updated:       2021-12-14
Ref:           https://rdap.arin.net/registry/ip/137.30.0.0

OrgName:       University of New Orleans
OrgId:         UNO
Address:       Computer Research Building
Address:       Lakefront Campus
City:          New Orleans
StateProv:     LA
PostalCode:    70148
Country:       US
RegDate:       1988-08-17
Updated:       2015-10-23
Ref:           https://rdap.arin.net/registry/entity/UNO

OrgAbuseHandle: IEN2-ARIN
OrgAbuseName:   IT Enterprise Networking
OrgAbusePhone:  +1-504-280-1198
```

```
OrgAbuseEmail: hachoi@uno.edu
OrgAbuseRef: https://rdap.arin.net/registry/entity/IEN2-ARIN

OrgNOCHandle: IEN2-ARIN
OrgNOCName: IT Enterprise Networking
OrgNOCPhone: +1-504-280-1198
OrgNOCEmail: hachoi@uno.edu
OrgNOCRef: https://rdap.arin.net/registry/entity/IEN2-ARIN

OrgTechHandle: IEN2-ARIN
OrgTechName: IT Enterprise Networking
OrgTechPhone: +1-504-280-1198
OrgTechEmail: hachoi@uno.edu
OrgTechRef: https://rdap.arin.net/registry/entity/IEN2-ARIN

RAbuseHandle: IEN2-ARIN
RAbuseName: IT Enterprise Networking
RAbusePhone: +1-504-280-1198
RAbuseEmail: hachoi@uno.edu
RAbuseRef: https://rdap.arin.net/registry/entity/IEN2-ARIN

RNOCHandle: IEN2-ARIN
RNOCName: IT Enterprise Networking
RNOCPHONE: +1-504-280-1198
```

```
RNOCEmail: hachoi@uno.edu
RNOCRef: https://rdap.arin.net/registry/entity/IEN2-ARIN

RTechHandle: IEN2-ARIN
RTechName: IT Enterprise Networking
RTechPhone: +1-504-280-1198
RTechEmail: hachoi@uno.edu
RTechRef: https://rdap.arin.net/registry/entity/IEN2-ARIN
```

```
#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# █
```

[illegible]

```

└─# xxd rhino4.jpg | head
00000000: ffd8 ffe0 0010 4a46 4946 0001 0001 012c .....JFIF.....,
00000010: 012c 0000 ff0e 0017 552d 4c65 6164 2053 .,.....U-Lead S
00000020: 7973 7465 6d73 2c20 496e 632e 00ff db00 ystems, Inc.....
00000030: 8400 422d 323a 3229 423a 363a 4b47 424f ..B-2:2)B:6:KGB0
00000040: 64a7 6c64 5b5b 64cc 929a 79a7 f2d5 fefa d.ld[d ... y.....
00000050: eed5 eae5 ffff ffff ffff ffff e5ea ffff .....
00000060: ffff ffff ffff ffff ffff ffff ffff ffff .....
00000070: ffff 0147 4b4b 6457 64c4 6c6c c4ff ffea ... GKKdWd.ll....
00000080: ffff ffff ffff ffff ffff ffff ffff ffff .....
00000090: ffff ffff ffff ffff ffff ffff ffff ffff .....

```

```

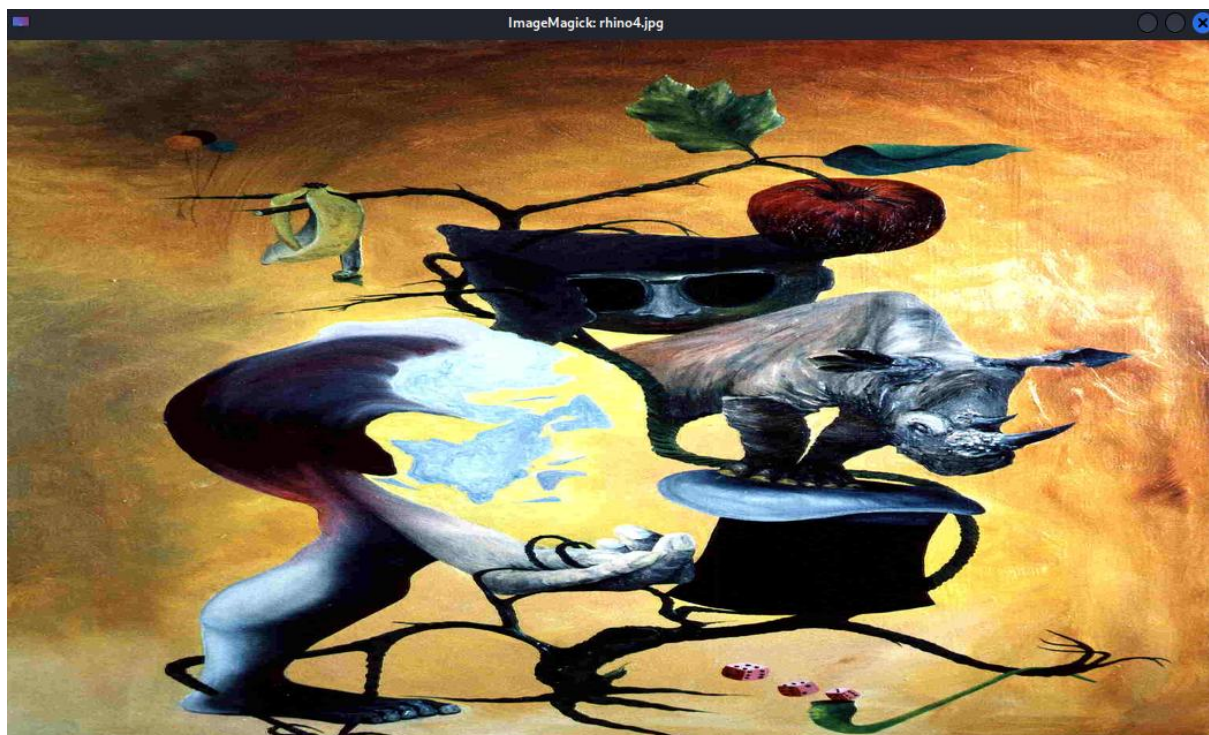
└─(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
└─# xxd rhino4.jpg | tail
000255d0: 5002 1eb4 74ef 4006 78a4 a002 8a00 5c73 P... t.@.x.....\s
000255e0: 498a 402d 1400 5028 0146 33d2 9d4f 700c I.@- .. P(.F3..Op.
000255f0: 0f4a 6145 f4a1 a043 1a3c 534a 62a2 c55c .JaE ... C.<Sjb..\
00025600: 4db8 e693 69a4 02e2 8e68 1851 8a00 3145 M... i....h.Q..1E
00025610: 0210 f141 a004 c525 300c 518a 0031 da8c ... A ... %0.Q..1..
00025620: 6280 1314 8450 2109 c500 d031 e0d3 4e3d b....P!....1..N=
00025630: 2810 840a 6b0a 0637 6e29 00cd 300c e28c ( ... k..7n)..0...
00025640: 0c53 0023 d29b 8a00 4346 2980 7a53 a900 .S.#....CF).zS..
00025650: bde9 40c5 2000 3de9 76d0 01b6 976d 0018 ..@. .=.v....m..
00025660: e714 d340 00ff d9 ... @ ...

```

```

└─(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
└─# █

```



File Edit View Go Capture Analyze S

tcp.stream eq 2

No. Time

40 2004-04-28 21:08:36.34278

41 2004-04-28 21:08:36.34366

42 2004-04-28 21:08:36.34368

43 2004-04-28 21:08:36.34438

44 2004-04-28 21:08:36.34536

46 2004-04-28 21:08:36.35237

48 2004-04-28 21:08:36.47306

217 2004-04-28 21:08:44.18929

218 2004-04-28 21:08:44.19859

219 2004-04-28 21:08:44.22125

220 2004-04-28 21:08:44.22243

221 2004-04-28 21:08:44.22265

222 2004-04-28 21:08:44.25212

223 2004-04-28 21:08:44.25332

224 2004-04-28 21:08:44.25355

Frame 217: Packet, 488 bytes on wire (capture time: 0.000000 seconds) (interface: eth0)
 Ethernet II, Src: Apple_cc:57:92:1a:2b:2c, Dst: 08:00:0e:1c:3b:50
 Internet Protocol Version 4, Src: 137.30.123.234, Dst: 137.30.120.37
 Transmission Control Protocol, Src Port: 49206, Dst Port: 80, Seq: 312, Len: 0
 Hypertext Transfer Protocol
 GET /~gnome/rhino5.gif HTTP/1.1
 Accept: image/gif, image/x-bitmap, image/jpeg, image/pjpeg, application/vnd.ms-excel, application/vnd.ms-powerpoint, application/msword, application/x-shockwave-flash, */*
 Accept-Language: en-us
 User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; .NET CLR 1.1.4322)
 Host: www.cs.uno.edu
 Connection: Keep-Alive

Connection: Keep-Alive
 Content-Type: image/gif

GIF89a.....fff333.....!Nthis art is in the public domain. Kevin Hughes, k
 evinh@eit.com, September 1995.!.K...#...j.3S....i.7N.v..t..7"E.nXm/...5
 FP...x...l.(N.@..V...U...;
 GET /~gnome/rhino5.gif HTTP/1.1
 Accept: image/gif, image/x-bitmap, image/jpeg, image/pjpeg, application/vnd.ms-excel, ap
 plication/vnd.ms-powerpoint, application/msword, application/x-shockwave-flash, */*
 Accept-Language: en-us
 User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; .NET CLR 1.1.4322)
 Host: www.cs.uno.edu
 Connection: Keep-Alive

HTTP/1.1 200 OK
 Date: Wed, 28 Apr 2004 21:07:34 GMT
 Server: Apache/1.3.29 (Unix)
 Last-Modified: Wed, 28 Apr 2004 21:07:34 GMT
 ETag: W/"19b1e7-14c91-40901d9c"
 Accept-Ranges: bytes
 Content-Length: 85137
 Keep-Alive: timeout=15, max=99
 Connection: Keep-Alive
 Content-Type: image/gif

GIF89a.....fff333.....!Nthis art is in the public domain. Kevin Hughes, k
 evinh@eit.com, September 1995.!.K...#...j.3S....i.7N.v..t..7"E.nXm/...5
 FP...x...l.(N.@..V...U...;
 GET /~gnome/rhino5.gif HTTP/1.1
 Accept: image/gif, image/x-bitmap, image/jpeg, image/pjpeg, application/vnd.ms-excel, ap
 plication/vnd.ms-powerpoint, application/msword, application/x-shockwave-flash, */*
 Accept-Language: en-us
 User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; .NET CLR 1.1.4322)
 Host: www.cs.uno.edu
 Connection: Keep-Alive

Find: Entire conversation (86 kB) Show as ASCII No delta times Stream 2 Case sensitive Find Next

311 2004-04-28 21:08:44.377797Z 137.30.123.234 137.30.120.37 TCP

312 2004-04-28 21:08:44.378087Z 137.30.120.37 137.30.123.234 HTTP

313 2004-04-28 21:08:44.498636Z 137.30.123.234 137.30.120.37 TCP

320 2004-04-28 21:08:53.574938Z 137.30.123.234 137.30.120.37 TCP

323 2004-04-28 21:08:53.575584Z 137.30.120.37 137.30.123.234 TCP

[Request in frame: 217]
 [Time since request: 188.793000 milliseconds]
 [Request URI: /~gnome/rhino5.gif]
 [Full request URI: http://www.cs.uno.edu/~gnome/rhino5.gif]
 File Data: 85137 bytes

▼ Compuserve GIF, Version: GIF89a
 Version: GIF89a
 Screen width: 400
 Screen height: 275

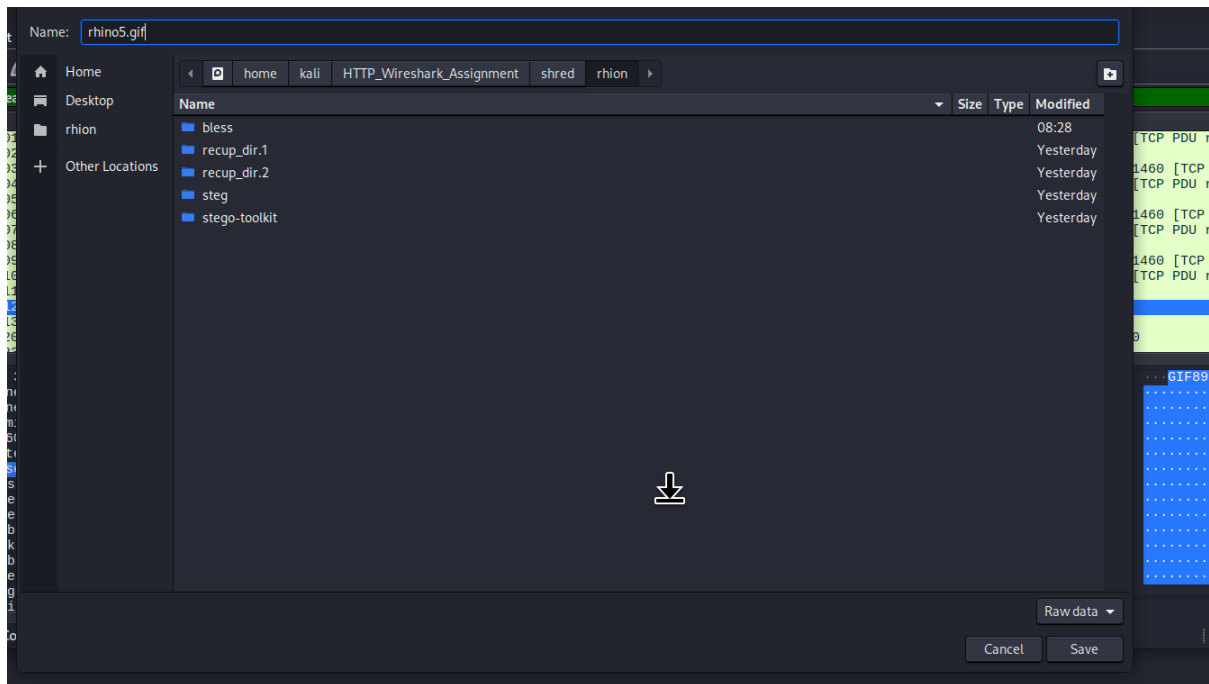
► Global settings: (Global color table present) (7 bits per color) (7 bits per pixel)
 Background color index: 0
 Global color map [...]: 000000ffffffbfdf2fbfdeef7f9eae7e9d6eff1dfebeddbfbfdebf9f

► Extension: Graphics Control
 ► Image
 Trailer (End of the GIF stream)

Last-Modified: Wed, 28 Apr 2004 21:07:34 GMT\r\n
 ETag: W/"19b1e7-14c91-40901d9c"\r\n
 Accept-Ranges: bytes\r\n
 Content-Length: 85137\r\n
 Keep-Alive: timeout=15, max=99\r\n
 Connection: Keep-Alive\r\n
 Content-Type: image/gif\r\n
 \r\n

[Request in frame: 217]
 [Time since request: 188.793000 milliseconds]
 [Request URI: /~gnome/rhino5.gif]
 [Full request URI: http://www.cs.uno.edu/~gnome/rhino5.gif]
 File Data: 85137 bytes

▼ Compuserve GIF, Version: GIF89a
 Version: GIF89a
 Screen width: 400



```
# xxd rhino5.gif | head
00000000: 4749 4638 3961 9001 1301 e600 0000 0000  GIF89a.....
00000010: ffff fffb fdf2 fbfd eef7 f9ea e7e9 d6ef  ....
00000020: f1df ebed dbfb fdeb f9fb e9eb eddd e7e9  ....
00000030: d9f9 fbeb fbfd e7f9 fbe5 f9fb e7df e0cf  ....
00000040: fdff e7fb fbe5 d3d3 c1fb fbe7 f9f9 e5e9  ....
00000050: e9d7 dbdb cafd fdeb fbfb e9f9 f9e7 eeee  ....
00000060: ddeb ebda e6e6 d5f6 f6e5 fbfb ebf9 f9e9  ....
00000070: f7f7 e7fd fdee f3f3 e4eb ebdd e7e7 d9fb  ....
00000080: fbed f9f9 ebf7 f7e9 efef e1ce cdbc eeef  ....
00000090: dae5 e4d2 e1e0 ced7 d6c5 f4f3 e1f2 f1e0  ....
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# xxd rhino5.gif | tail
00014c00: 5011 0a24 dfe3 1878 31a4 23e1 f902 31aa  P.. $... x1.# ... 1.
00014c10: 8025 29c1 6b66 b263 6058 36c1 1555 adec  .%).kf.c`X6.. U..
00014c20: 612b 7841 1c98 9114 8f75 0308 3179 981c  a+xA.....u.. 1y..
00014c30: 86c0 8b6f 40ee 4883 4a05 2adf f027 bb94  ... o@.H.J.*.. ' ..
00014c40: 524a 60e2 ca31 e592 4c32 ecad 2b9d 9112  RJ` .. 1.. L2 .. + ...
00014c50: 19f2 5025 1491 c198 e240 d35a 82e5 06b4  .. P%.....@.Z....
00014c60: 9430 2a76 e18f 5cf2 7095 ad2d 850c cd03  .0*v.. \.p.. -....
00014c70: 8a8a 60d2 170a 32a6 0cf0 8c53 4f24 888b  .. ` ... 2....S0$..
00014c80: 6ca1 932e f3b3 4c2f dee2 8631 8029 1000  l.....L/ ... 1.)..
00014c90: 3b                                     ;
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
#
```



http.content_type[0:5] == "image"							
No.	Time	Source	Destination	Protocol	Length	Info	
23	2004-04-28 21:08:31.150253Z	64.233.167.104	137.30.123.234	HTTP	1484	[TCP Out-Of-Order] HTTP/1.1 200 OK (GIF89a)	
37	2004-04-28 21:08:36.085585Z	137.30.120.37	137.30.123.234	HTTP	484	HTTP/1.1 200 OK (GIF89a)	
45	2004-04-28 21:08:36.347258Z	137.30.120.37	137.30.123.234	HTTP	646	HTTP/1.1 200 OK (GIF89a)	
46	2004-04-28 21:08:36.352379Z	137.30.120.37	137.30.123.234	HTTP	553	HTTP/1.1 200 OK (GIF89a)	
215	2004-04-28 21:08:38.274614Z	137.30.120.37	137.30.123.234	HTTP	273	HTTP/1.1 200 OK (JPEG JFIF image)	
312	2004-04-28 21:08:44.378087Z	137.30.120.37	137.30.123.234	HTTP	675	HTTP/1.1 200 OK (GIF89a)	

Internet Protocol Version 4, Src: 64.233.167.104, Dst: 137.30.123.234		0000	00 03 93 cc 57 92 00 0d	ed 41 a8 40 08 00 45 16
Transmission Control Protocol, Src Port: 80, Dst Port: 2024, Seq: 10226, Ack: 653		0010	05 be c3 4f 00 00 31 06	d3 80 40 e9 a7 68 89 1e
[7 Reassembled TCP Segments (8764 bytes): #13(1430), #14(1430), #16(1430), #18(1430)]		0020	7b ea 00 50 07 e8 44 63	cb f6 5b 1e c4 38 50 16
Hypertext Transfer Protocol		0030	0b 68 9c 1d 00 00 8e ab	fd c3 10 51 34 60 78 97
HTTP/1.1 200 OK\r\n		0040	1d b4 c5 67 1c 19 62 28	60 82 7c 9d 02 c0 81 a8
Content-Type: image/gif\r\n		0050	0a b8 20 8f 1d c8 d3 43	95 4a 20 80 98 93 47 fa
Last-Modified: Mon, 22 Mar 2004 23:04:23 GMT\r\n		0060	fe 0f 03 08 22 10 7c c0	03 5c 9c d8 a9 4c e0 3e
Expires: Sun, 17 Jan 2038 19:14:07 GMT\r\n		0070	40 6e ba 73 28 e3 ee 14	1e 3e 83 1b 16 fc 66 38
Server: GWS/2.1\r\n		0080	f0 c1 35 28 fc e8 b8 00	b8 c2 08 26 15 6c 10 eb
Content-Length: 8558\r\n		0090	26 4a 2f 5d 59 06 51 a8	81 85 ce 23 c3 60 bc a3
Date: Wed, 28 Apr 2004 21:08:29 GMT\r\n		00a0	62 b7 22 a9 a1 0f 80 b3	a2 b7 9f ab 00 65 c7 55
\r\n		00b0	16 e3 77 31 40 90 a1 4d	7a eb 95 dc f6 b6 91 3e
[Request in frame: 11]		00c0	2d d8 09 1a 91 65 9a 56	a0 1e a3 8b a9 a9 ae ba
[Time since request: 195.154000 milliseconds]		00d0	17 3b a8 ad 16 8e 2c 0c	91 24 6c 11 4e 39 c5 83
[Request URI: /images/logo.gif]				
[Full request URI: http://www.google.com/images/logo.gif]				

Packet (1484 bytes)	Reassembled TCP (8764 bytes)
---------------------	------------------------------

```
\r\n
[Request in frame: 11]
[Time since request: 195.154000 milliseconds]
[Request URI: /images/logo.gif]
[Full request URI: http://www.google.com/images/logo.gif]
File Data: 8558 bytes
Compuserve GIF, Version: GIF89a
Version: GIF89a
Screen width: 276
Screen height: 110
Global settings: (Global color table present) (7 bits per color) (8 bits per pixel)
Background color index: 0
Global color map [...]: f7f7f7fffbffe7e7d6d3d6efebefcecbcead1400dedbde1845ad184
Image
Trailer (End of the GIF stream)
```

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# openssl s_client -connect imap.gmail.com:993 -crlf
Connecting to 172.253.143.108
CONNECTED(00000003)
depth=2 C=US, O=Google Trust Services LLC, CN=GTS Root R1
verify return:1
depth=1 C=US, O=Google Trust Services, CN=WR2
verify return:1
depth=0 CN=imap.gmail.com
verify return:1
—
Certificate chain
 0 s:CN=imap.gmail.com
  i:C=US, O=Google Trust Services, CN=WR2
  a:PKEY: EC, (prime256v1); sigalg: sha256WithRSAEncryption
  v:NotBefore: Aug 10 08:39:01 2026 GMT; NotAfter: Nov  2 08:39:00 2026 GMT
 1 s:C=US, O=Google Trust Services, CN=WR2
  i:C=US, O=Google Trust Services LLC, CN=GTS Root R1
  a:PKEY: RSA, 2048 (bit); sigalg: sha256WithRSAEncryption
  v:NotBefore: Dec 13 09:00:00 2023 GMT; NotAfter: Feb 20 14:00:00 2029 GMT
 2 s:C=US, O=Google Trust Services LLC, CN=GTS Root R1
  i:C=BE, O=GlobalSign nv-sa, OU=Root CA, CN=GlobalSign Root CA
  a:PKEY: RSA, 4096 (bit); sigalg: sha256WithRSAEncryption
  v:NotBefore: Jun 19 00:00:42 2020 GMT; NotAfter: Jan 28 00:00:42 2028 GMT
—
Server certificate
```

```
TLS session ticket lifetime hint: 172799 (seconds)
TLS session ticket:
0000 - 02 9b fa 6f 3c 2b 6f f0-97 a9 94 bc a1 d8 24 0c ... o<+o.....$.
0010 - 2b 1f 63 2e aa 6e 32 09-f4 01 3e e2 89 82 f4 2f +.c..n2 ...>.../
0020 - 39 5a 08 e4 37 83 ea c9-22 52 2e dc 78 58 69 49 9Z..7 ... "R..xXiI
0030 - 0a d4 be 39 66 13 62 c9-5b 22 fd ec 14 2a ee d3 ... 9f.b.[" ... *..
0040 - 7f 1a 82 b7 3e 40 9c 62-61 2a 33 ca 9c 45 7a 27 ....>@.ba*3..Ez'
0050 - 33 08 04 ad 12 c0 e9 d1-f5 a5 6b 56 29 91 7b 17 3.....kV).{.
0060 - 66 96 35 65 fb f5 08 e2-24 94 2a f7 db 11 9c 47 f.5e....$.*....G
0070 - bc 31 af 32 f7 1f c4 4b-ce 49 e3 15 25 a3 38 8e .1.2 ... K.I...%.8.
0080 - cb 8e 9b 95 63 c2 c8 ba-04 c1 3c 15 66 00 e5 58 ....c.....<.f..X
0090 - c7 03 9c 53 c8 e3 04 c2-ca 46 af 72 36 bd b9 71 ... S.....F.r6..q
00a0 - 11 e8 d5 9a 2c 95 bc 82-b2 96 fe d9 8a 5d cb 2b .....,.....].+
00b0 - 7c aa 26 63 f3 03 19 81-12 7f 01 b4 47 94 79 cd |.8c.....G.y.
00c0 - 15 03 4f f1 14 e2 a5 6a-b4 9d 1f cd 7e 7c 2b 58 ..0....j.....~|+X
00d0 - c6 ae 0f 9c 40 8f 0d 62-e9 42 37 07 b7 cb b0 4e ...@...b.B7...N
00e0 - 94 3a 61 69 11 42 52 55-cf b7 1c 31 fe a5 7b b6 ..:ai.BRU ... 1..{.

Start Time: 1788962483
Timeout : 7200 (sec)
Verify return code: 0 (ok)
Extended master secret: no
Max Early Data: 0

read R BLOCK
* OK Gimap ready for requests from 102.176.101.64 5b1f17b1804b1-49d2549263fmb206572805e9
```

http contains ".exe"

No.	Time	Source	Destination	Protocol	Length	Info
10	2004-04-28 21:15:48.234997Z	137.30.123.234	64.233.167.99	HTTP	592	GET /search?hl=en&ie=UTF-8&oe=UTF-8&q=rhino.exe HTTP/1.1
28	2004-04-28 21:15:48.725700Z	64.233.167.99	137.30.123.234	HTTP	1112	HTTP/1.1 200 OK (text/html)
59	2004-04-28 21:15:55.113910Z	137.30.123.234	216.239.51.99	HTTP	653	GET /groups?q=rhino.exe&hl=en&lr=&ie=UTF-8&oe=UTF-8&sa=N&tab=wg HTTP/1.1
85	2004-04-28 21:16:02.002505Z	137.30.123.234	216.239.51.99	HTTP	719	GET /groups?q=rhino.exe&hl=en&lr=&ie=UTF-8&oe=UTF-8&selm=MPG.159359f528ff3d2...
110	2004-04-28 21:16:23.343061Z	137.30.123.234	137.30.120.37	HTTP	447	GET /-gnome/rhino.exe HTTP/1.1

```
HTTP/1.1 200 OK
Date: Wed, 28 Apr 2004 21:15:13 GMT
Server: Apache/1.3.29 (Unix)
Last-Modified: Wed, 28 Apr 2004 21:15:13 GMT
ETag: w/"19b286-23a00-40901fb9"
Accept-Ranges: bytes
Content-Length: 145920
Keep-Alive: timeout=15, max=100
Connection: Keep-Alive
Content-Type: application/octet-stream
```

```
$.....M.....3.....~.....3.....Rich.....
.....PE.L.....};
.....(.....hp.
.....0.....X.....
.....text.....^.....data.....0.....@.....rsrc
.....hp.....r.....@.....@.....};H.....};U.....};b.....};b.....};l.....};w.....};
.....};
.....ADVAPI32.dll.KERNEL32.dll.NTDLL.DLL.RPCRT4.dll.ole32.dll.SETUPAPI.dll.U
SER32.dll
```

Entire conversation (146 kB) Show as ASCII No delta times Stream 3

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[Save as...](#)
[Back](#)
[× Close](#)

No.	Time	Source	Destination	Protocol	Length	Info
265	2004-04-28 21:16:24.208235Z	137.39.123.234	137.39.120.37	TCP	54	2163 - 80 [ACK] Seq=394 Ack=138681 Win=64240 Len=0
266	2004-04-28 21:16:24.209183Z	137.39.120.37	137.39.123.234	TCP	1514	80 - 2163 [ACK] Seq=138681 Ack=394 Win=49247 Len=1460 [TCP PDU reassembl
267	2004-04-28 21:16:24.209414Z	137.39.123.234	137.39.120.37	TCP	54	2163 - 80 [ACK] Seq=394 Ack=140141 Win=64240 Len=0
268	2004-04-28 21:16:24.210397Z	137.39.120.37	137.39.123.234	TCP	1514	80 - 2163 [ACK] Seq=140141 Ack=394 Win=49247 Len=1460 [TCP PDU reassembl
269	2004-04-28 21:16:24.211623Z	137.39.120.37	137.39.123.234	TCP	1514	80 - 2163 [ACK] Seq=141681 Ack=394 Win=49247 Len=1460 [TCP PDU reassembl
270	2004-04-28 21:16:24.211954Z	137.39.123.234	137.39.120.37	TCP	54	2163 - 80 [ACK] Seq=394 Ack=143061 Win=64240 Len=0
271	2004-04-28 21:16:24.212833Z	137.39.120.37	137.39.123.234	TCP	1514	80 - 2163 [ACK] Seq=143061 Ack=394 Win=49247 Len=1460 [TCP PDU reassembl
272	2004-04-28 21:16:24.213956Z	137.39.123.234	137.39.120.37	TCP	54	2163 - 80 [ACK] Seq=394 Ack=144521 Win=64240 Len=0
273	2004-04-28 21:16:24.214047Z	137.39.120.37	137.39.123.234	TCP	1514	80 - 2163 [ACK] Seq=144521 Ack=394 Win=49247 Len=1460 [TCP PDU reassembl
274	2004-04-28 21:16:24.214222Z	137.39.120.37	137.39.123.234	HTTP	302	HTTP/1.1 200 OK
275	2004-04-28 21:16:24.214286Z	137.39.123.234	137.39.120.37	TCP	54	2163 - 80 [ACK] Seq=394 Ack=146229 Win=64240 Len=0
276	2004-04-28 21:16:24.376455Z	137.39.120.37	137.39.123.234	TCP	60	80 - 2163 [FIN, ACK] Seq=146229 Ack=394 Win=49247 Len=0
279	2004-04-28 21:16:24.376590Z	137.39.123.234	137.39.120.37	TCP	54	2163 - 80 [ACK] Seq=394 Ack=146230 Win=64240 Len=0
283	2004-04-28 21:16:43.376726Z	137.39.123.234	137.39.120.37	TCP	54	2163 - 80 [FIN, ACK] Seq=394 Ack=146230 Win=64240 Len=0
284	2004-04-28 21:16:43.377382Z	137.39.120.37	137.39.123.234	TCP	60	80 - 2163 [ACK] Seq=146230 Ack=395 Win=49247 Len=0

Frame 274: Packet, 302 bytes on wire (2416 bits), 302 bytes captured (2416 bits) on 0

Ethernet II, Src: Oracle 72:6f:0b (08:00:20:72:6f:0b), Dst: Apple cc:57:92 (08:00:93:00:00:00), Src: 137.39.120.37, Dst: 137.39.123.234

Internet Protocol Version 4, Src: 137.39.120.37, Dst: 137.39.123.234

Transmission Control Protocol, Src Port: 80, Dst Port: 2163, Seq: 145981, Ack: 394, Len: 145920

[144 Reassembled TCP Segments (146228 bytes) #: 1112(1460), #113(1460), #115(1460), #117(1460), #119(1460), #121(1460), #123(1460), #125(1460), #127(1460), #129(1460), #131(1460), #133(1460), #135(1460), #137(1460), #139(1460), #141(1460), #143(1460), #145(1460), #147(1460), #149(1460), #151(1460), #153(1460), #155(1460), #157(1460), #159(1460), #161(1460), #163(1460), #165(1460), #167(1460), #169(1460), #171(1460), #173(1460), #175(1460), #177(1460), #179(1460), #181(1460), #183(1460), #185(1460), #187(1460), #189(1460), #191(1460), #193(1460), #195(1460), #197(1460), #199(1460), #201(1460), #203(1460), #205(1460), #207(1460), #209(1460), #211(1460), #213(1460), #215(1460), #217(1460), #219(1460), #221(1460), #223(1460), #225(1460), #227(1460), #229(1460), #231(1460), #233(1460), #235(1460), #237(1460), #239(1460), #241(1460), #243(1460), #245(1460), #247(1460), #249(1460), #251(1460), #253(1460), #255(1460), #257(1460), #259(1460), #261(1460), #263(1460), #265(1460), #267(1460), 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#2

```
(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# ls -l rhino.exe
-rw-r--r-- 1 root root 145920 Sep  9 19:13 rhino.exe

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# file rhino.exe
rhino.exe: PE32 executable for MS Windows 4.00 (console), Intel i386, 3 sections

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# wine rhino.exe
008c:err:vulkan:vulkan_init_once Failed to load libvulkan.so.1
008c:err:xrandr:vulkan_init_once Failed to load libvulkan.so.1
00c0:err:hid:stop_polling_device could not find poll entry matching device 0x7f3025f0 fd
00c0:err:hid:stop_polling_device could not find poll entry matching device 0x7f32c9b0 fd
00c0:err:hid:stop_polling_device could not find poll entry matching device 0x7f3354f0 fd
00c0:err:hid:stop_polling_device could not find poll entry matching device 0x7f346d40 fd
00c0:err:hid:stop_polling_device could not find poll entry matching device 0x7f34fe10 fd
00c0:err:hid:stop_polling_device could not find poll entry matching device 0x7f356610 fd
00c0:err:hid:stop_polling_device could not find poll entry matching device 0x7f35d5a0 fd
00c0:err:hid:stop_polling_device could not find poll entry matching device 0x7f3646c0 fd
00c0:err:hid:stop_polling_device could not find poll entry matching device 0x7f36b630 fd
00c0:err:hid:stop_polling_device could not find poll entry matching device 0x7f3729a0 fd
00c0:err:hid:stop_polling_device could not find poll entry matching device 0x7f378e30 fd

Microsoft DiskPart version 1.0
Copyright (C) 1999-2001 Microsoft Corporation.
On computer: KALI
0024:err:ole:com_get_class_object class {4fb6bb00-3347-11d0-b40a-00aa005ff586} not registered
0024:err:ole:create_server_class {4fb6bb00-3347-11d0-b40a-00aa005ff586} not registered
0024:err:ole:com_get_class_object no class object {4fb6bb00-3347-11d0-b40a-00aa005ff586} could be created for context 0x15

The disk management services could not complete the operation.

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
# md5sum rhino.exe
d62d9989535c4c8db14e50b58c9f25a0 rhino.exe

(root@kali)-[/home/kali/HTTP_Wireshark_Assignment/shred/rhion]
#
```



ANY.RUN

<https://app.any.run/tasks/62d9989535c4c8db14e50b58c9f25a0>

D62D9989535C4C8DB14E50B5...

Interactive malware hunting service. Live testing of most type of threats in any environments. No installation and no waiting necessary.



AI Overview

d62d9989535c4c8db14e50b58c9f25a0 is the MD5 cryptographic hash value for a specific file from a classic digital forensics challenge. [National Institute of Standards and Technology](#)

Context & Details

- The File:** This hash identifies an executable associated with the **Microsoft DiskPart Application (version 1.0)**.
- The Origin:** It is a well-known artifact from the **DFRWS 2005 Forensic Rodeo Challenge** (specifically featured in digital forensics exercises like the "Rhino" case). [National Institute of Standards and Technology](#) +1

Investigators and students often encounter this specific hash when analyzing simulated

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Partner

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23K

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diskp.exe File Summary

File Format: [EXE](#)
Category: Disk Wiper, File Management
Latest Software: Hard Drive Eraser Disk Wiper
Software Version: none
Author: Altertech Systems

Name: diskp.exe [Request](#)
Bytes: 145920
SHA-1: 91619dffa529908c8c81189a0c5ec95ade6f9992
MD5: d62d9989535c4c8db14e50b58c9f25a0
CRC32: 0aedd09

Diskp.exe

1

Select OS Version

Macintosh

2

Application

Hard Drive Eraser Disk Wiper